CM22009: Machine Learning

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Hierarchical clustering

12/11/2025

Announcements

Topics Covered Today

- Hierarchical Clustering
  - Agglomerative
  - Divisive

Limitations of K-means

- Unknown number of clusters K
- Hard cluster assignment
- Sensitive to initialization
- Constrained Cluster Shapes
- Sensitive to outliers

Hierarchical Clustering

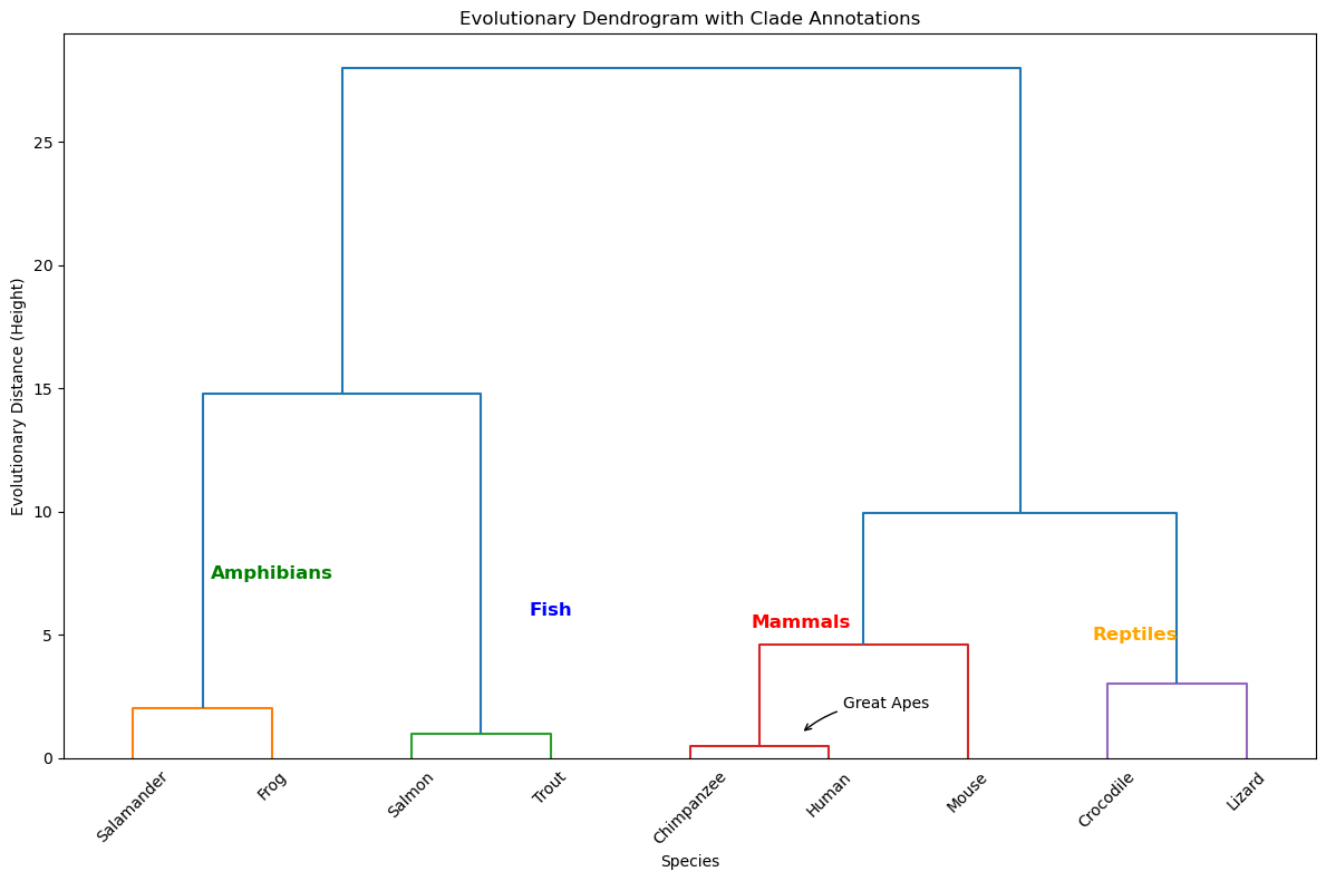
- Creates a hierarchy of clusters
- Each datapoint belongs to one or more clusters (soft assignment)
- Each cluster belongs to another cluster, or it includes all the data
- Two Approaches:

    I. **Agglomerative (Bottom-up):** It starts by treating each datapoint as a single cluster. Then pairs of clusters are merged as one moves up the hierarchy. It's the more common type.

    II. **Divisive (Top-down):** Starts with all observations in a single cluster and splits iteratively as one moves down the hierarchy

Hierarchical Clustering Examples

- Species Hierarchy

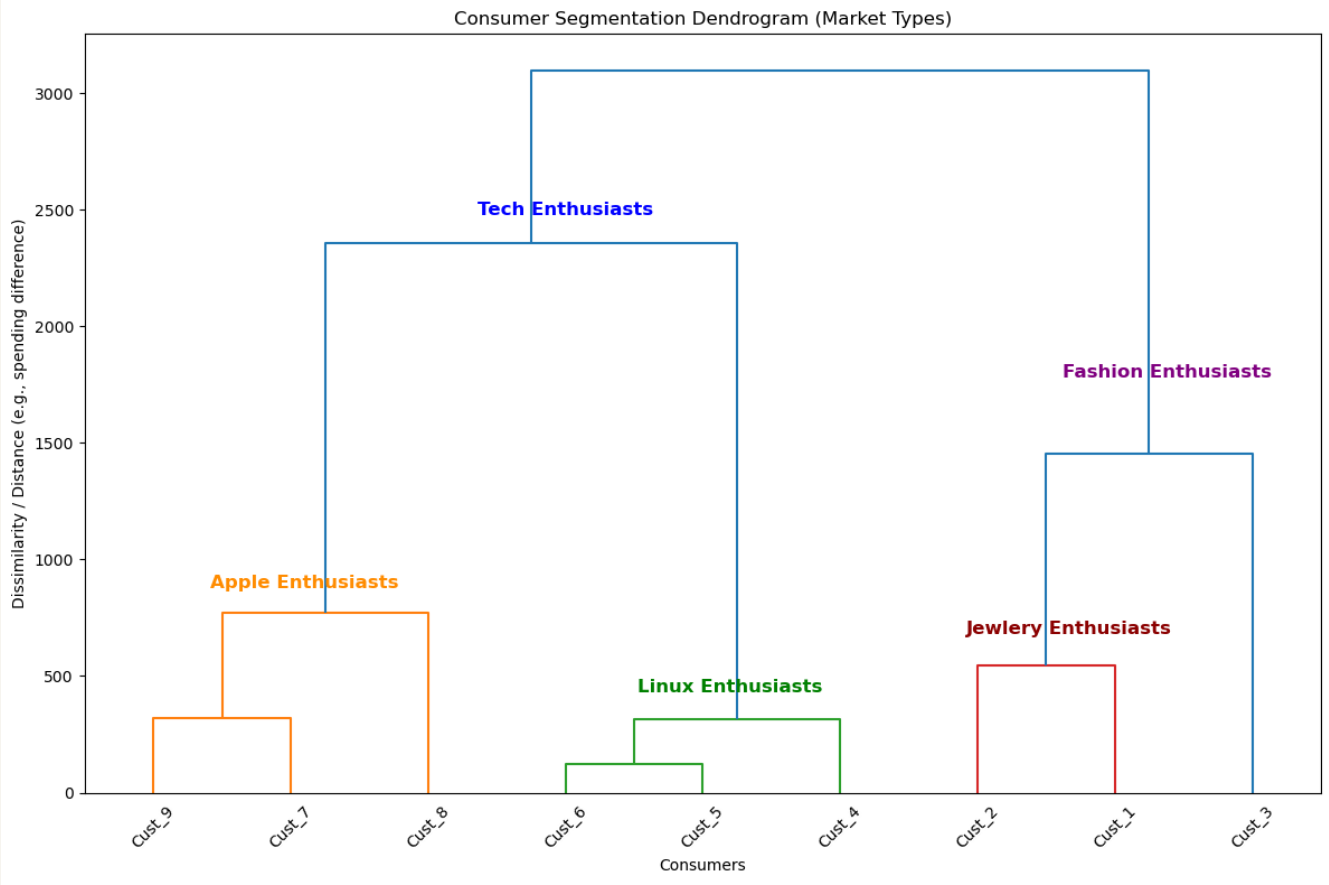


Results represented by a dendrogram

Hierarchical Clustering Examples

- Consumer Hierarchy

Divisive



Agglomerative

Agglomerative (Bottom-up)

1. Starts with N clusters, each containing one datapoint
2. **Finding Closest Clusters:** At each step, the algorithm identifies and merges two clusters that are closest to each other. This closeness is usually measured through various distance metrics like Euclidean or Manhattan distance.
3. **Linkage Criteria:** Determines which distance to use between sets of observations. Common methods include:
   - **Single Linkage:** Distance between the **closest** members of two clusters.
   - **Complete Linkage:** Distance between the **farthest** members.
   - **Average Linkage: Mean** distance between members.
4. **Repeating Steps:** These steps are repeated until all data points are clustered into a single group (or a stop criterion is met).

Divisive (Top-down)

- Initially, all data points are considered as part of one big cluster.
- **Split the clusters in two:**
  - Initialise a new cluster with the datapoint that has the highest distance from the mean. Separate the data in the two clusters based on the distances from the data and the new means. (Sensitive to Outliers)
  - You can do that with K-means with $K = 2$ (Sensitive to Initialization)
  - Many other approaches appear in the literature, depending on the problem
- **Repeat** the Splitting step for each cluster until every cluster has only one datapoint or a termination criterion is met, e.g. K clusters
- The Divisive approach is less common than the agglomerative.

Example – Cluster Cities based on Geographic Distance

City	Latitude	Longitude
London	51.5074	-0.1278
Bristol	51.4545	-2.5879
Bath	51.3811	-2.3570
Paris	48.8566	2.3522
Lyon	45.7640	4.8357
Berlin	52.5200	13.4050
Stuttgart	48.7758	9.1829
Frankfurt	50.1109	8.6821
Bremen	53.0793	8.8017
Edinburgh	55.9533	-3.1883
Glasgow	55.8642	-4.2514
Toulouse	43.6045	1.4442
Bordeaux	44.8378	-0.5792

Example – Cluster Cities based on Geographic Distance

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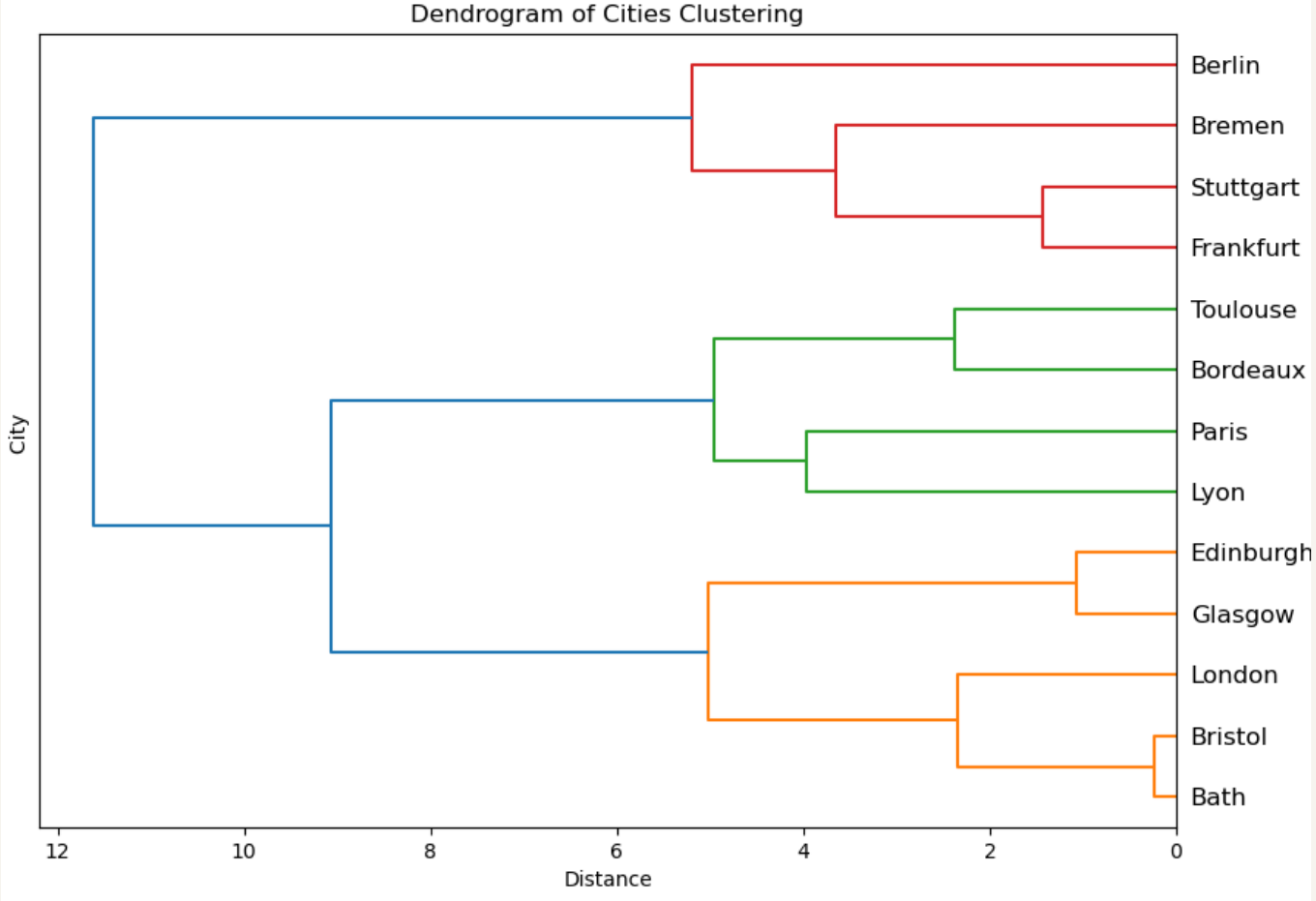


Agglomerative Clustering

Evaluate Cluster Distances

	London	Bristol	Bath	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux
London	0.0	2.5	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7
Bristol	2.5	0.0	0.2	5.6	9.4	16.0	12.1	11.3	11.5	4.5	4.7	8.8	6.9
Bath	2.2	0.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.6	4.9	8.7	6.8
Paris	3.6	5.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0
Lyon	7.6	9.4	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5
Berlin	13.6	16.0	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0
Stuttgart	9.7	12.1	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5
Frankfurt	8.9	11.3	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7
Bremen	9.1	11.5	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5
Edinburgh	5.4	4.5	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4
Glasgow	6.0	4.7	4.9	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6
Toulouse	8.1	8.8	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4
Bordeaux	6.7	6.9	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0

Agglomerative Clustering

Evaluate Cluster Distances

Identify the two closest datapoints

	London	Bristol	Bath	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux
London	0.0	2.5	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7
Bristol	2.5	0.0	0.2	5.6	9.4	16.0	12.1	11.3	11.5	4.5	4.7	8.8	6.9
Bath	2.2	0.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.6	4.9	8.7	6.8
Paris	3.6	5.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0
Lyon	7.6	9.4	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5
Berlin	13.6	16.0	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0
Stuttgart	9.7	12.1	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5
Frankfurt	8.9	11.3	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7
Bremen	9.1	11.5	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5
Edinburgh	5.4	4.5	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4
Glasgow	6.0	4.7	4.9	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6
Toulouse	8.1	8.8	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4
Bordeaux	6.7	6.9	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0

Agglomerative Clustering

Merge the Clusters: Single Linkage

Min

	London	Bristol	Bath	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux
London	0.0	2.5	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7
Bristol	2.5	0.0	0.2	5.6	9.4	16.0	12.1	11.3	11.5	4.5	4.7	8.8	6.9
Bath	2.2	0.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.6	4.9	8.7	6.8
Paris	3.6	5.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0
Lyon	7.6	9.4	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5
Berlin	13.6	16.0	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0
Stuttgart	9.7	12.1	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5
Frankfurt	8.9	11.3	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7
Bremen	9.1	11.5	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5
Edinburgh	5.4	4.5	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4
Glasgow	6.0	4.7	4.9	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6
Toulouse	8.1	8.8	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4
Bordeaux	6.7	6.9	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0

Min

Agglomerative Clustering

Merge the Clusters: Complete Linkage

Max

	London	Bristol	Bath	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux
London	0.0	2.5	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7
Bristol	2.5	0.0	0.2	5.6	9.4	16.0	12.1	11.3	11.5	4.5	4.7	8.8	6.9
Bath	2.2	0.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.6	4.9	8.7	6.8
Paris	3.6	5.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0
Lyon	7.6	9.4	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5
Berlin	13.6	16.0	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0
Stuttgart	9.7	12.1	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5
Frankfurt	8.9	11.3	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7
Bremen	9.1	11.5	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5
Edinburgh	5.4	4.5	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4
Glasgow	6.0	4.7	4.9	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6
Toulouse	8.1	8.8	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4
Bordeaux	6.7	6.9	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0

Max

Agglomerative Clustering

Merge the Clusters: Average Linkage

Avg

	London	Bristol	Bath	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux
London	0.0	2.5	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7
Bristol	2.5	0.0	0.2	5.6	9.4	16.0	12.1	11.3	11.5	4.5	4.7	8.8	6.9
Bath	2.2	0.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.6	4.9	8.7	6.8
Paris	3.6	5.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0
Lyon	7.6	9.4	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5
Berlin	13.6	16.0	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0
Stuttgart	9.7	12.1	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5
Frankfurt	8.9	11.3	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7
Bremen	9.1	11.5	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5
Edinburgh	5.4	4.5	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4
Glasgow	6.0	4.7	4.9	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6
Toulouse	8.1	8.8	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4
Bordeaux	6.7	6.9	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0

Avg

Agglomerative Clustering

Merge the Clusters: Single Linkage

	London	[Bristol, Bath]	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux
London	0.0	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7
[Bristol, Bath]	2.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.5	4.7	8.7	6.8
Paris	3.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0
Lyon	7.6	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5
Berlin	13.6	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0
Stuttgart	9.7	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5
Frankfurt	8.9	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7
Bremen	9.1	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5
Edinburgh	5.4	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4
Glasgow	6.0	4.7	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6
Toulouse	8.1	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4
Bordeaux	6.7	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0

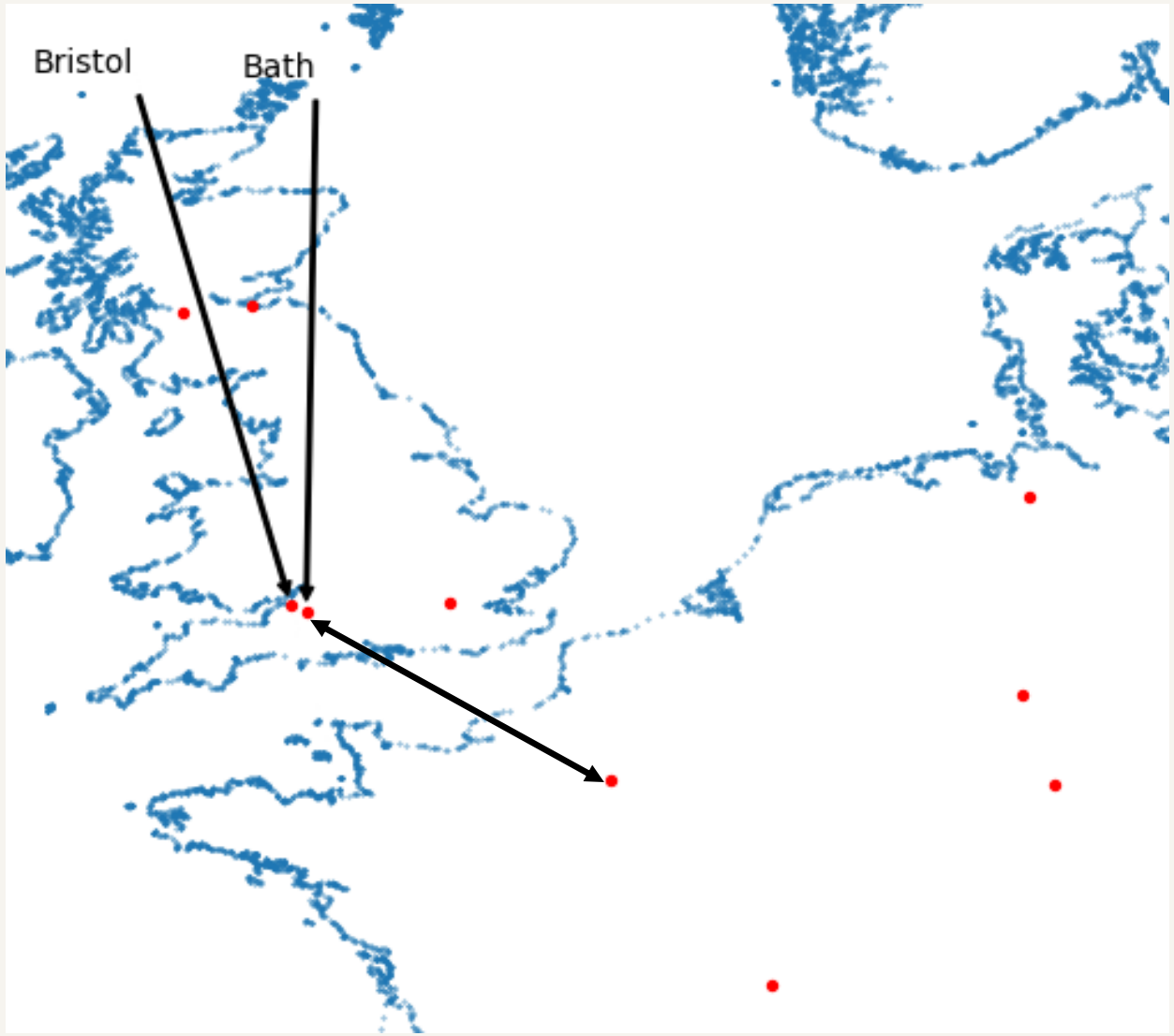
Agglomerative Clustering

First Group
{Bristol, Bath}

Agglomerative Clustering

First Group
{Bristol, Bath}

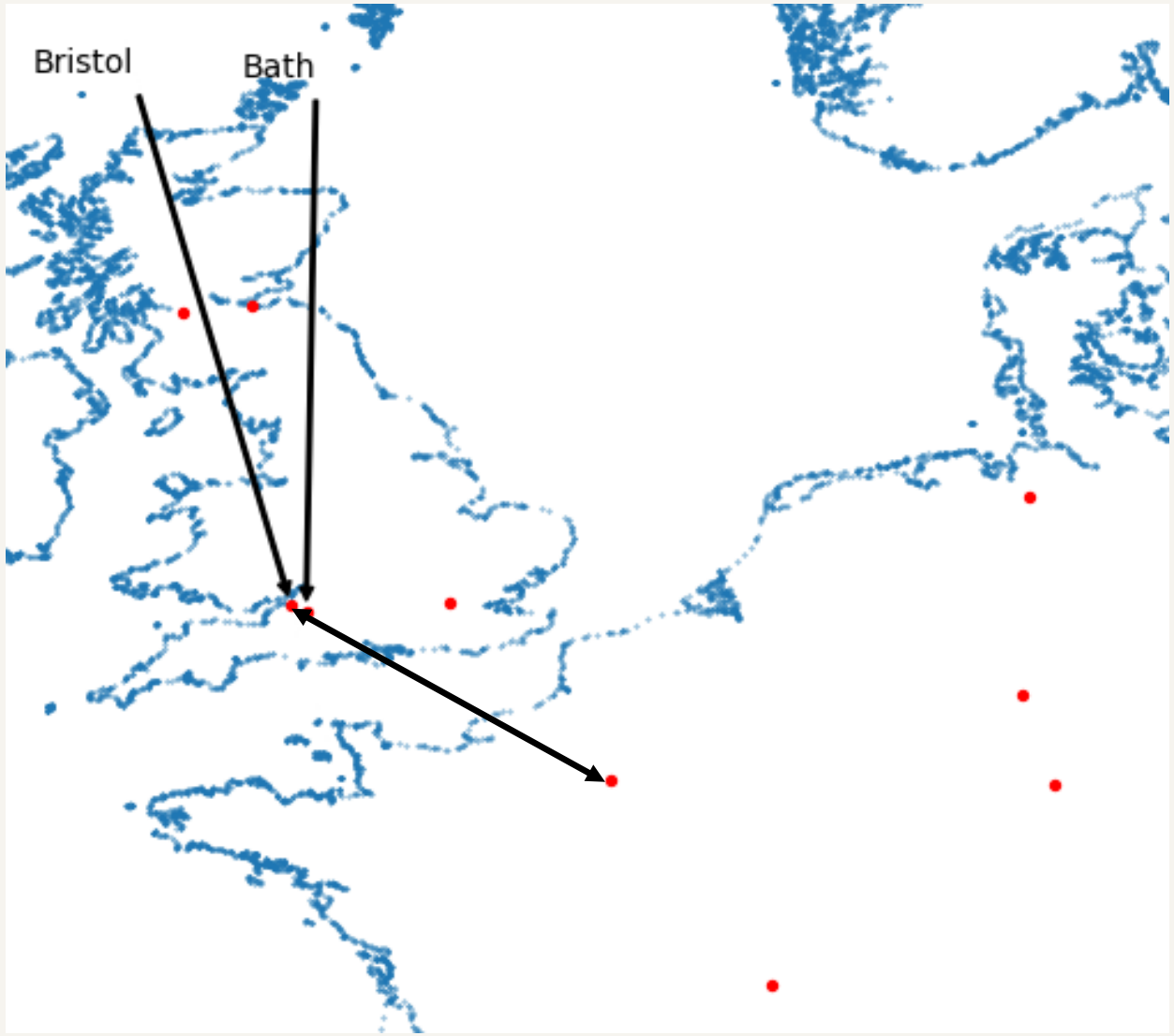
Single Linkage:
Distance between the group and other nodes, is the minimum distance from any member of the group and the other nodes (Bath)



Agglomerative Clustering

First Group
{Bristol, Bath}

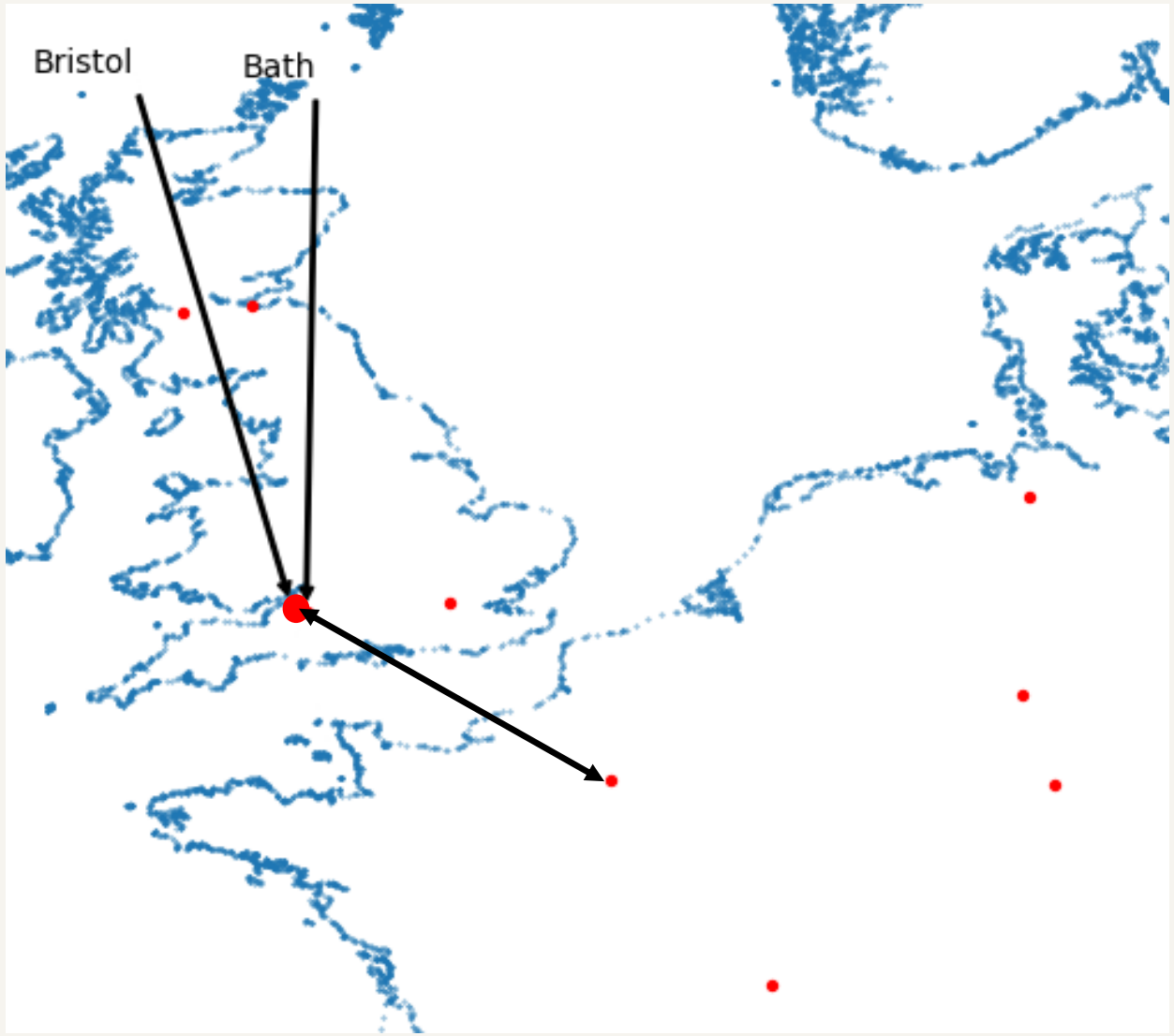
Complete Linkage:
Distance between the group and other nodes, is the maximum distance from any member of the group and the other nodes (Bath)



Agglomerative Clustering

First Group
{Bristol, Bath}

Average Linkage:
Distance between the group and other nodes, is the minimum distance from any member of the group and the other nodes (Bath)



Agglomerative Clustering

Merge the Clusters: Single Linkage

	London	[Bristol, Bath]	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux
London	0.0	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7
[Bristol, Bath]	2.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.5	4.7	8.7	6.8
Paris	3.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0
Lyon	7.6	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5
Berlin	13.6	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0
Stuttgart	9.7	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5
Frankfurt	8.9	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7
Bremen	9.1	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5
Edinburgh	5.4	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4
Glasgow	6.0	4.7	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6
Toulouse	8.1	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4
Bordeaux	6.7	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0

Agglomerative Clustering

Merge the Clusters: Single Linkage

	London	[Bristol, Bath]	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux
London	0.0	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7
[Bristol, Bath]	2.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.5	4.7	8.7	6.8
Paris	3.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0
Lyon	7.6	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5
Berlin	13.6	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0
Stuttgart	9.7	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5
Frankfurt	8.9	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7
Bremen	9.1	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5
Edinburgh	5.4	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4
Glasgow	6.0	4.7	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6
Toulouse	8.1	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4
Bordeaux	6.7	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0

Agglomerative Clustering

Merge the Clusters: Single Linkage

	London	[Bristol, Bath]	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux
London	0.0	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7
[Bristol, Bath]	2.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.5	4.7	8.7	6.8
Paris	3.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0
Lyon	7.6	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5
Berlin	13.6	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0
Stuttgart	9.7	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5
Frankfurt	8.9	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7
Bremen	9.1	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5
Edinburgh	5.4	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4
Glasgow	6.0	4.7	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6
Toulouse	8.1	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4
Bordeaux	6.7	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0

Agglomerative Clustering

Merge the Clusters: Single Linkage

	London	[Bristol, Bath]	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	[Edinburgh, Glasgow]	Toulouse	Bordeaux
London	0.0	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	8.1	6.7
[Bristol, Bath]	2.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.5	8.7	6.8
Paris	3.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	5.3	5.0
Lyon	7.6	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	4.0	5.5
Berlin	13.6	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	14.9	16.0
Stuttgart	9.7	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	9.3	10.5
Frankfurt	8.9	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	9.7	10.7
Bremen	9.1	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	12.0	12.5
[Edinburgh, Glasgow]	5.4	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	13.2	11.4
Toulouse	8.1	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	0.0	2.4
Bordeaux	6.7	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	2.4	0.0

Agglomerative Clustering

Merge the Clusters: Single Linkage

	London	[Bristol, Bath]	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	[Edinburgh, Glasgow]	Toulouse	Bordeaux
London	0.0	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	8.1	6.7
[Bristol, Bath]	2.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.5	8.7	6.8
Paris	3.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	5.3	5.0
Lyon	7.6	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	4.0	5.5
Berlin	13.6	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	14.9	16.0
Stuttgart	9.7	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	9.3	10.5
Frankfurt	8.9	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	9.7	10.7
Bremen	9.1	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	12.0	12.5
[Edinburgh, Glasgow]	5.4	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	13.2	11.4
Toulouse	8.1	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	0.0	2.4
Bordeaux	6.7	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	2.4	0.0

Agglomerative Clustering

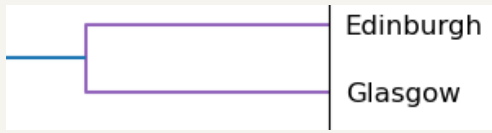
Merge the Clusters: Single Linkage

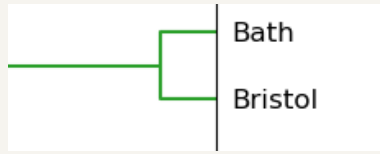
	London	[Bristol, Bath]	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	[Edinburgh, Glasgow]	Toulouse	Bordeaux
London	0.0	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	8.1	6.7
[Bristol, Bath]	2.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.5	8.7	6.8
Paris	3.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	5.3	5.0
Lyon	7.6	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	4.0	5.5
Berlin	13.6	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	14.9	16.0
Stuttgart	9.7	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	9.3	10.5
Frankfurt	8.9	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	9.7	10.7
Bremen	9.1	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	12.0	12.5
[Edinburgh, Glasgow]	5.4	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	13.2	11.4
Toulouse	8.1	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	0.0	2.4
Bordeaux	6.7	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	2.4	0.0

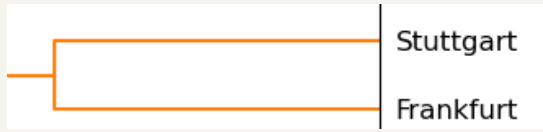
Agglomerative Clustering

Merge the Clusters: Single Linkage

	London	[Bristol, Bath]	Paris	Lyon	Berlin	[Frankfurt, Stuttgart]	Bremen	[Edinburgh, Glasgow]	Toulouse	Bordeaux
London	0.0	2.2	3.6	7.6	13.6	8.9	9.1	5.4	8.1	6.7
[Bristol, Bath]	2.2	0.0	5.3	9.1	15.8	11.1	11.3	4.5	8.7	6.8
Paris	3.6	5.3	0.0	4.0	11.6	6.5	7.7	9.0	5.3	5.0
Lyon	7.6	9.1	4.0	0.0	10.9	5.3	8.3	13.0	4.0	5.5
Berlin	13.6	15.8	11.6	10.9	0.0	5.3	4.6	16.9	14.9	16.0
[Frankfurt, Stuttgart]	8.9	11.1	6.5	5.3	5.3	0.0	3.0	13.2	9.3	10.5
Bremen	9.1	11.3	7.7	8.3	4.6	3.0	0.0	12.3	12.0	12.5
[Edinburgh, Glasgow]	5.4	4.6	9.0	13.0	16.9	13.2	12.3	0.0	13.2	11.4
Toulouse	8.1	8.7	5.3	4.0	14.9	9.3	12.0	13.2	0.0	2.4
Bordeaux	6.7	6.8	5.0	5.5	16.0	10.5	12.5	11.4	2.4	0.0



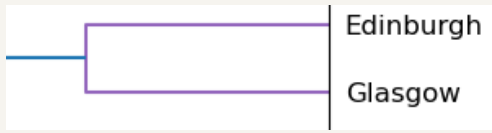


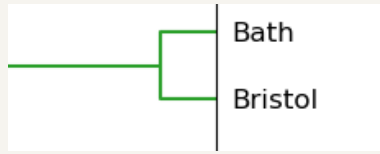


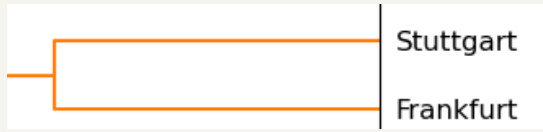
Agglomerative Clustering

Merge the Clusters: Single Linkage

	London	[Bristol, Bath]	Paris	Lyon	Berlin	[Frankfurt, Stuttgart]	Bremen	[Edinburgh, Glasgow]	Toulouse	Bordeaux
London	0.0	2.2	3.6	7.6	13.6	8.9	9.1	5.4	8.1	6.7
[Bristol, Bath]	2.2	0.0	5.3	9.1	15.8	11.1	11.3	4.5	8.7	6.8
Paris	3.6	5.3	0.0	4.0	11.6	6.5	7.7	9.0	5.3	5.0
Lyon	7.6	9.1	4.0	0.0	10.9	5.3	8.3	13.0	4.0	5.5
Berlin	13.6	15.8	11.6	10.9	0.0	5.3	4.6	16.9	14.9	16.0
[Frankfurt, Stuttgart]	8.9	11.1	6.5	5.3	5.3	0.0	3.0	13.2	9.3	10.5
Bremen	9.1	11.3	7.7	8.3	4.6	3.0	0.0	12.3	12.0	12.5
[Edinburgh, Glasgow]	5.4	4.6	9.0	13.0	16.9	13.2	12.3	0.0	13.2	11.4
Toulouse	8.1	8.7	5.3	4.0	14.9	9.3	12.0	13.2	0.0	2.4
Bordeaux	6.7	6.8	5.0	5.5	16.0	10.5	12.5	11.4	2.4	0.0



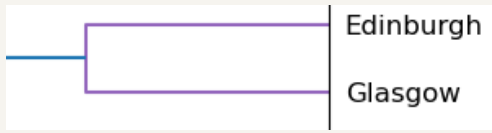


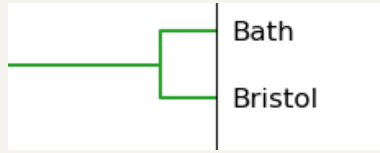


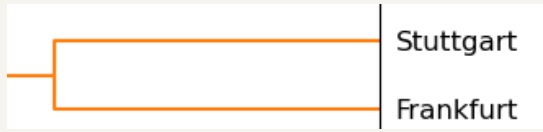
Agglomerative Clustering

Merge the Clusters: Single Linkage

	London	{Bristol, Bath}	Paris	Lyon	Berlin	{Frankfurt, Stuttgart}	Bremen	{Edinburgh, Glasgow}	Toulouse	Bordeaux
London	0.0	2.2	3.6	7.6	13.6	8.9	9.1	5.4	8.1	6.7
{Bristol, Bath}	2.2	0.0	5.3	9.1	15.8	11.1	11.3	4.5	8.7	6.8
Paris	3.6	5.3	0.0	4.0	11.6	6.5	7.7	9.0	5.3	5.0
Lyon	7.6	9.1	4.0	0.0	10.9	5.3	8.3	13.0	4.0	5.5
Berlin	13.6	15.8	11.6	10.9	0.0	5.3	4.6	16.9	14.9	16.0
{Frankfurt, Stuttgart}	8.9	11.1	6.5	5.3	5.3	0.0	3.0	13.2	9.3	10.5
Bremen	9.1	11.3	7.7	8.3	4.6	3.0	0.0	12.3	12.0	12.5
{Edinburgh, Glasgow}	5.4	4.6	9.0	13.0	16.9	13.2	12.3	0.0	13.2	11.4
Toulouse	8.1	8.7	5.3	4.0	14.9	9.3	12.0	13.2	0.0	2.4
Bordeaux	6.7	6.8	5.0	5.5	16.0	10.5	12.5	11.4	2.4	0.0



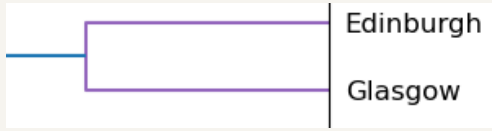


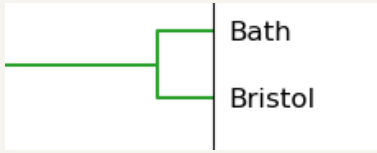


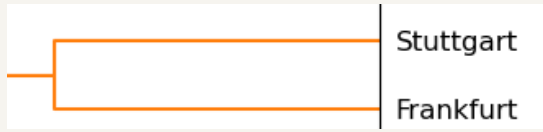
Agglomerative Clustering

Merge the Clusters: Single Linkage

	[London, [Bristol, Bath]]	Paris	Lyon	Berlin	[Frankfurt, Stuttgart]	Bremen	[Edinburgh, Glasgow]	Toulouse	Bordeaux
[London, [Bristol, Bath]]	0.0	3.6	7.6	13.6	8.9	9.1	4.5	8.1	6.7
Paris	3.6	0.0	4.0	11.6	6.5	7.7	9.0	5.3	5.0
Lyon	7.6	4.0	0.0	10.9	5.3	8.3	13.0	4.0	5.5
Berlin	13.6	11.6	10.9	0.0	5.3	4.6	16.9	14.9	16.0
[Frankfurt, Stuttgart]	8.9	6.5	5.3	5.3	0.0	3.0	13.2	9.3	10.5
Bremen	9.1	7.7	8.3	4.6	3.0	0.0	12.3	12.0	12.5
[Edinburgh, Glasgow]	4.6	9.0	13.0	16.9	13.2	12.3	0.0	13.2	11.4
Toulouse	8.1	5.3	4.0	14.9	9.3	12.0	13.2	0.0	2.4
Bordeaux	6.7	5.0	5.5	16.0	10.5	12.5	11.4	2.4	0.0



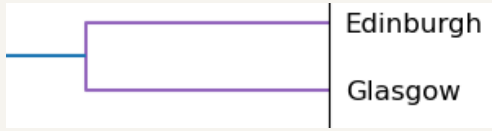


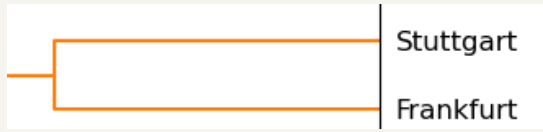


Agglomerative Clustering

Merge the Clusters: Single Linkage

	[London, [Bristol, Bath]]	Paris	Lyon	Berlin	[Frankfurt, Stuttgart]	Bremen	[Edinburgh, Glasgow]	Toulouse	Bordeaux
[London, [Bristol, Bath]]	0.0	3.6	7.6	13.6	8.9	9.1	4.5	8.1	6.7
Paris	3.6	0.0	4.0	11.6	6.5	7.7	9.0	5.3	5.0
Lyon	7.6	4.0	0.0	10.9	5.3	8.3	13.0	4.0	5.5
Berlin	13.6	11.6	10.9	0.0	5.3	4.6	16.9	14.9	16.0
[Frankfurt, Stuttgart]	8.9	6.5	5.3	5.3	0.0	3.0	13.2	9.3	10.5
Bremen	9.1	7.7	8.3	4.6	3.0	0.0	12.3	12.0	12.5
[Edinburgh, Glasgow]	4.6	9.0	13.0	16.9	13.2	12.3	0.0	13.2	11.4
Toulouse	8.1	5.3	4.0	14.9	9.3	12.0	13.2	0.0	2.4
Bordeaux	6.7	5.0	5.5	16.0	10.5	12.5	11.4	2.4	0.0

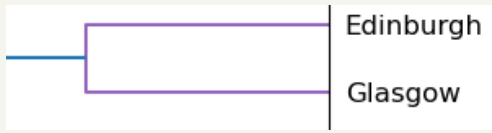


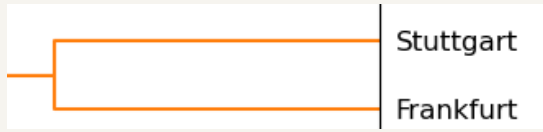


Agglomerative Clustering

Merge the Clusters: Single Linkage

	[London, [Bristol, Bath]]	Paris	Lyon	Berlin	[Frankfurt, Stuttgart]	Bremen	[Edinburgh, Glasgow]	Toulouse	Bordeaux
[London, [Bristol, Bath]]	0.0	3.6	7.6	13.6	8.9	9.1	4.5	8.1	6.7
Paris	3.6	0.0	4.0	11.6	6.5	7.7	9.0	5.3	5.0
Lyon	7.6	4.0	0.0	10.9	5.3	8.3	13.0	4.0	5.5
Berlin	13.6	11.6	10.9	0.0	5.3	4.6	16.9	14.9	16.0
[Frankfurt, Stuttgart]	8.9	6.5	5.3	5.3	0.0	3.0	13.2	9.3	10.5
Bremen	9.1	7.7	8.3	4.6	3.0	0.0	12.3	12.0	12.5
[Edinburgh, Glasgow]	4.6	9.0	13.0	16.9	13.2	12.3	0.0	13.2	11.4
Toulouse	8.1	5.3	4.0	14.9	9.3	12.0	13.2	0.0	2.4
Bordeaux	6.7	5.0	5.5	16.0	10.5	12.5	11.4	2.4	0.0

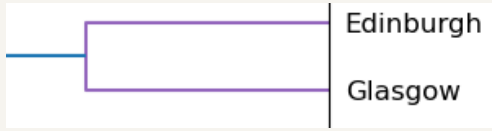


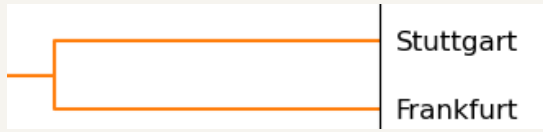


Agglomerative Clustering

Merge the Clusters: Single Linkage

	[London, [Bristol, Bath]]	Paris	Lyon	Berlin	[Frankfurt, Stuttgart]	Bremen	[Edinburgh, Glasgow]	Toulouse	Bordeaux
[London, [Bristol, Bath]]	0.0	3.6	7.6	13.6	8.9	9.1	4.5	8.1	6.7
Paris	3.6	0.0	4.0	11.6	6.5	7.7	9.0	5.3	5.0
Lyon	7.6	4.0	0.0	10.9	5.3	8.3	13.0	4.0	5.5
Berlin	13.6	11.6	10.9	0.0	5.3	4.6	16.9	14.9	16.0
[Frankfurt, Stuttgart]	8.9	6.5	5.3	5.3	0.0	3.0	13.2	9.3	10.5
Bremen	9.1	7.7	8.3	4.6	3.0	0.0	12.3	12.0	12.5
[Edinburgh, Glasgow]	4.6	9.0	13.0	16.9	13.2	12.3	0.0	13.2	11.4
Toulouse	8.1	5.3	4.0	14.9	9.3	12.0	13.2	0.0	2.4
Bordeaux	6.7	5.0	5.5	16.0	10.5	12.5	11.4	2.4	0.0

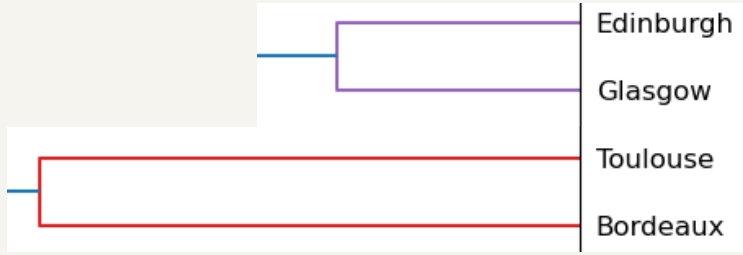


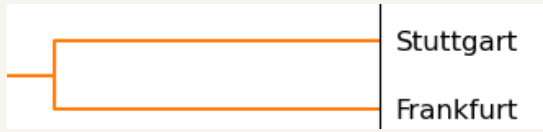


Agglomerative Clustering

Merge the Clusters: Single Linkage

	[London, [Bristol, Bath]]	Paris	Lyon	Berlin	[Frankfurt, Stuttgart]	Bremen	[Edinburgh, Glasgow]	[Toulouse, Bordeaux]
[London, [Bristol, Bath]]	0.0	3.6	7.6	13.6	8.9	9.1	4.5	6.7
Paris	3.6	0.0	4.0	11.6	6.5	7.7	9.0	5.0
Lyon	7.6	4.0	0.0	10.9	5.3	8.3	13.0	4.0
Berlin	13.6	11.6	10.9	0.0	5.3	4.6	16.9	14.9
[Frankfurt, Stuttgart]	8.9	6.5	5.3	5.3	0.0	3.0	13.2	9.3
Bremen	9.1	7.7	8.3	4.6	3.0	0.0	12.3	12.0
[Edinburgh, Glasgow]	4.6	9.0	13.0	16.9	13.2	12.3	0.0	11.4
[Toulouse, Bordeaux]	6.7	5.0	4.0	14.9	9.3	12.0	11.4	0.0

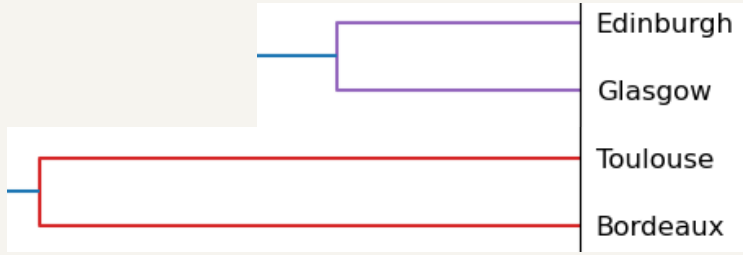


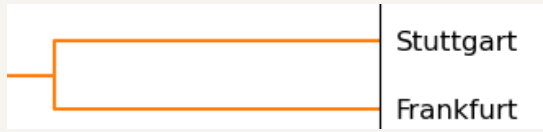


Agglomerative Clustering

Merge the Clusters: Single Linkage

	[London, [Bristol, Bath]]	Paris	Lyon	Berlin	[Frankfurt, Stuttgart]	Bremen	[Edinburgh, Glasgow]	[Toulouse, Bordeaux]
[London, [Bristol, Bath]]	0.0	3.6	7.6	13.6	8.9	9.1	4.5	6.7
Paris	3.6	0.0	4.0	11.6	6.5	7.7	9.0	5.0
Lyon	7.6	4.0	0.0	10.9	5.3	8.3	13.0	4.0
Berlin	13.6	11.6	10.9	0.0	5.3	4.6	16.9	14.9
[Frankfurt, Stuttgart]	8.9	6.5	5.3	5.3	0.0	3.0	13.2	9.3
Bremen	9.1	7.7	8.3	4.6	3.0	0.0	12.3	12.0
[Edinburgh, Glasgow]	4.6	9.0	13.0	16.9	13.2	12.3	0.0	11.4
[Toulouse, Bordeaux]	6.7	5.0	4.0	14.9	9.3	12.0	11.4	0.0

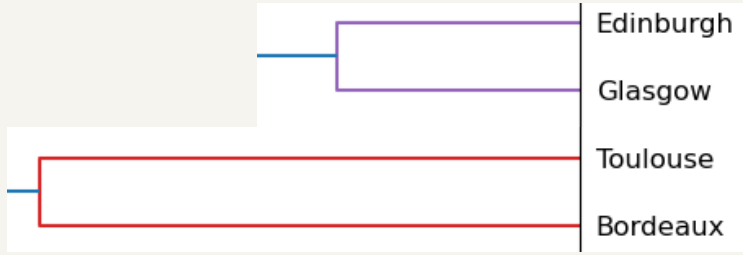


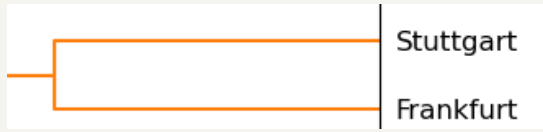


Agglomerative Clustering

Merge the Clusters: Single Linkage

	[London, [Bristol, Bath]]	Paris	Lyon	Berlin	[Frankfurt, Stuttgart]	Bremen	[Edinburgh, Glasgow]	[Toulouse, Bordeaux]
[London, [Bristol, Bath]]	0.0	3.6	7.6	13.6	8.9	9.1	4.5	6.7
Paris	3.6	0.0	4.0	11.6	6.5	7.7	9.0	5.0
Lyon	7.6	4.0	0.0	10.9	5.3	8.3	13.0	4.0
Berlin	13.6	11.6	10.9	0.0	5.3	4.6	16.9	14.9
[Frankfurt, Stuttgart]	8.9	6.5	5.3	5.3	0.0	3.0	13.2	9.3
Bremen	9.1	7.7	8.3	4.6	3.0	0.0	12.3	12.0
[Edinburgh, Glasgow]	4.6	9.0	13.0	16.9	13.2	12.3	0.0	11.4
[Toulouse, Bordeaux]	6.7	5.0	4.0	14.9	9.3	12.0	11.4	0.0

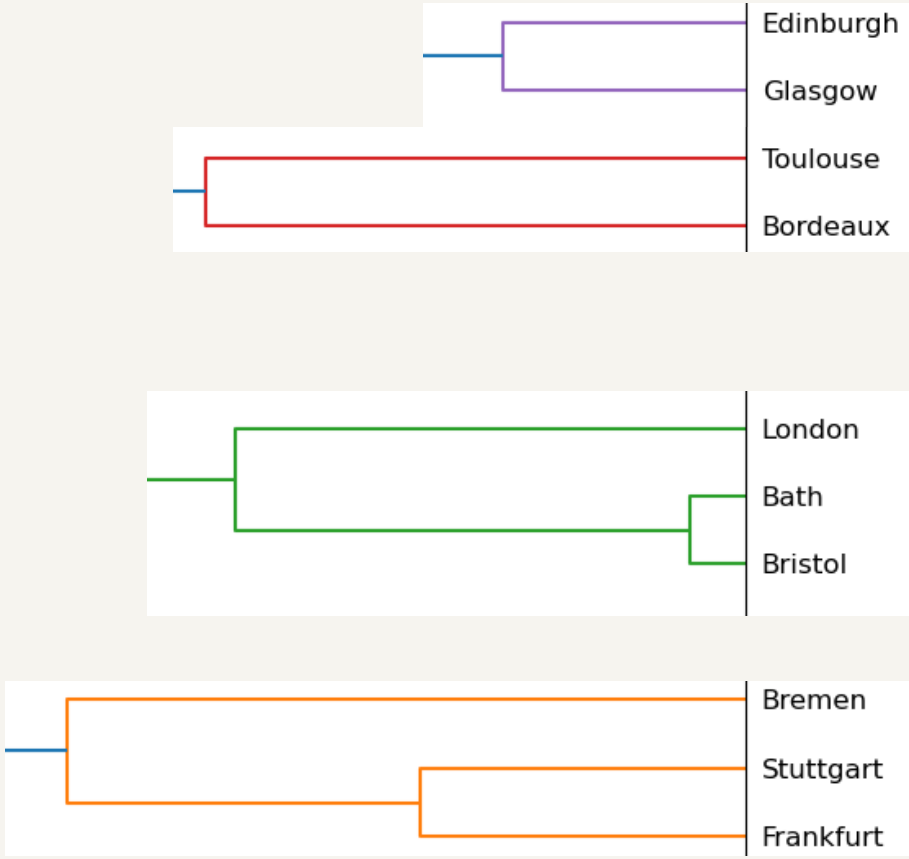




Agglomerative Clustering

Merge the Clusters: Single Linkage

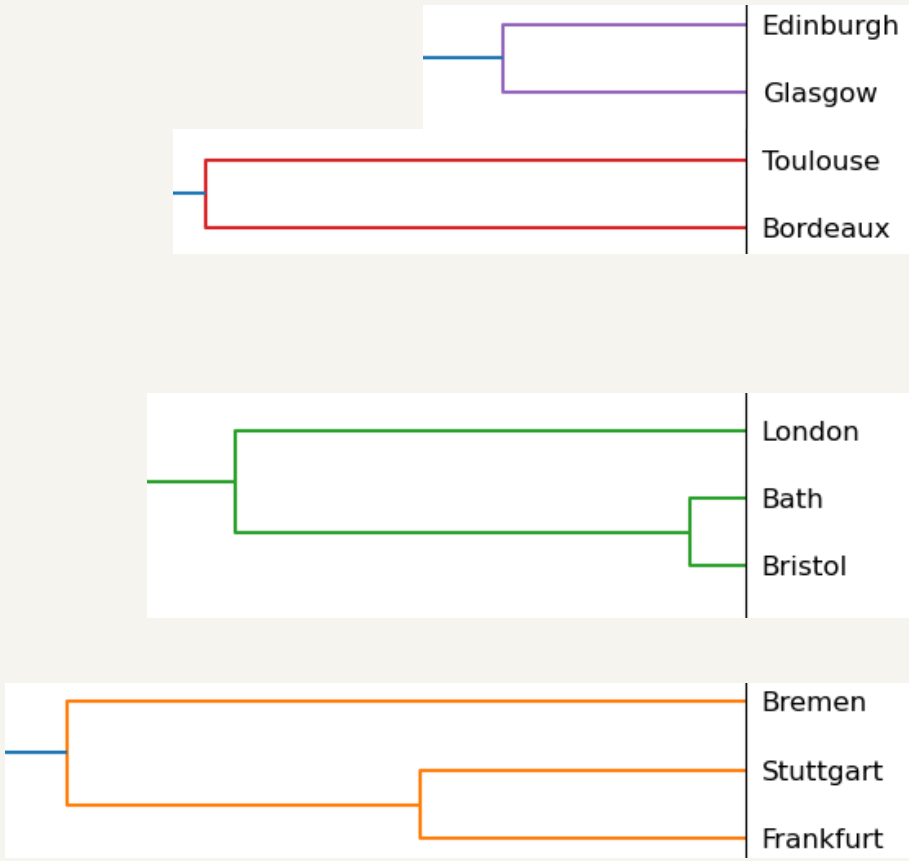
	[London, [Bristol, Bath]]	Paris	Lyon	Berlin	[[Frankfurt, Stuttgart], Bremen]	[Edinburgh, Glasgow]	[Toulouse, Bordeaux]
[London, [Bristol, Bath]]	0.0	3.6	7.6	13.6	8.9	4.5	6.7
Paris	3.6	0.0	4.0	11.6	6.5	9.0	5.0
Lyon	7.6	4.0	0.0	10.9	5.3	13.0	4.0
Berlin	13.6	11.6	10.9	0.0	4.6	16.9	14.9
[[Frankfurt, Stuttgart], Bremen]	8.9	6.5	5.3	4.6	0.0	12.3	9.3
[Edinburgh, Glasgow]	4.6	9.0	13.0	16.9	12.3	0.0	11.4
[Toulouse, Bordeaux]	6.7	5.0	4.0	14.9	9.3	11.4	0.0



Agglomerative Clustering

Merge the Clusters: Single Linkage

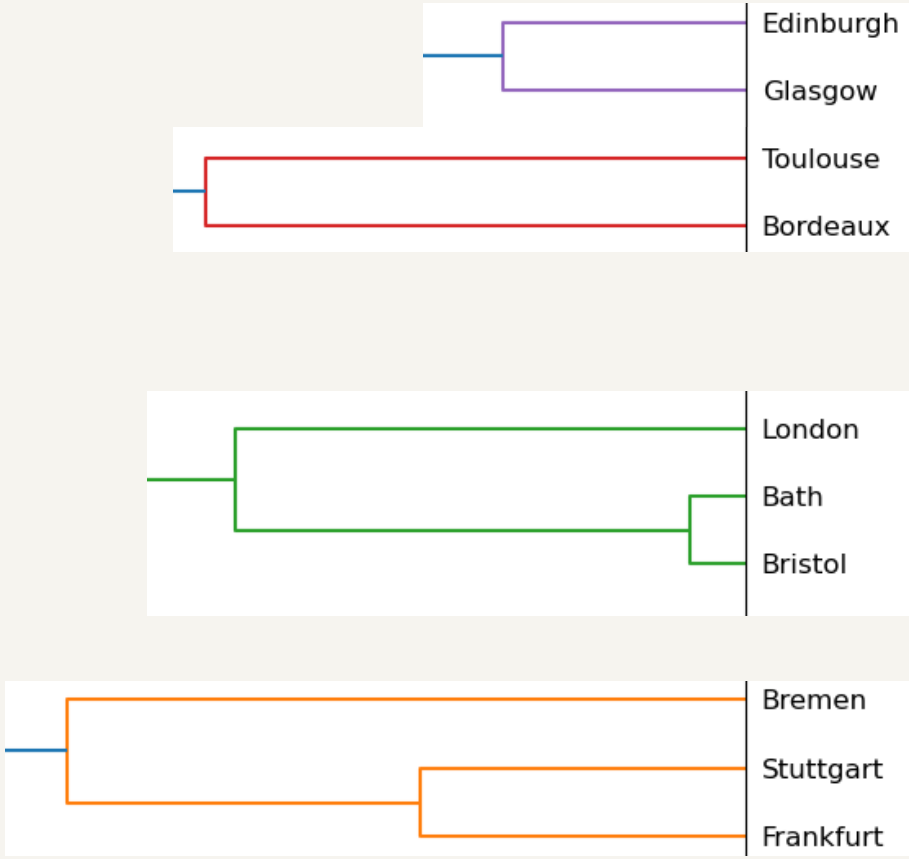
	[London, [Bristol, Bath]]	Paris	Lyon	Berlin	[[Frankfurt, Stuttgart], Bremen]	[Edinburgh, Glasgow]	[Toulouse, Bordeaux]
[London, [Bristol, Bath]]	0.0	3.6	7.6	13.6	8.9	4.5	6.7
Paris	3.6	0.0	4.0	11.6	6.5	9.0	5.0
Lyon	7.6	4.0	0.0	10.9	5.3	13.0	4.0
Berlin	13.6	11.6	10.9	0.0	4.6	16.9	14.9
[[Frankfurt, Stuttgart], Bremen]	8.9	6.5	5.3	4.6	0.0	12.3	9.3
[Edinburgh, Glasgow]	4.6	9.0	13.0	16.9	12.3	0.0	11.4
[Toulouse, Bordeaux]	6.7	5.0	4.0	14.9	9.3	11.4	0.0



Agglomerative Clustering

Merge the Clusters: Single Linkage

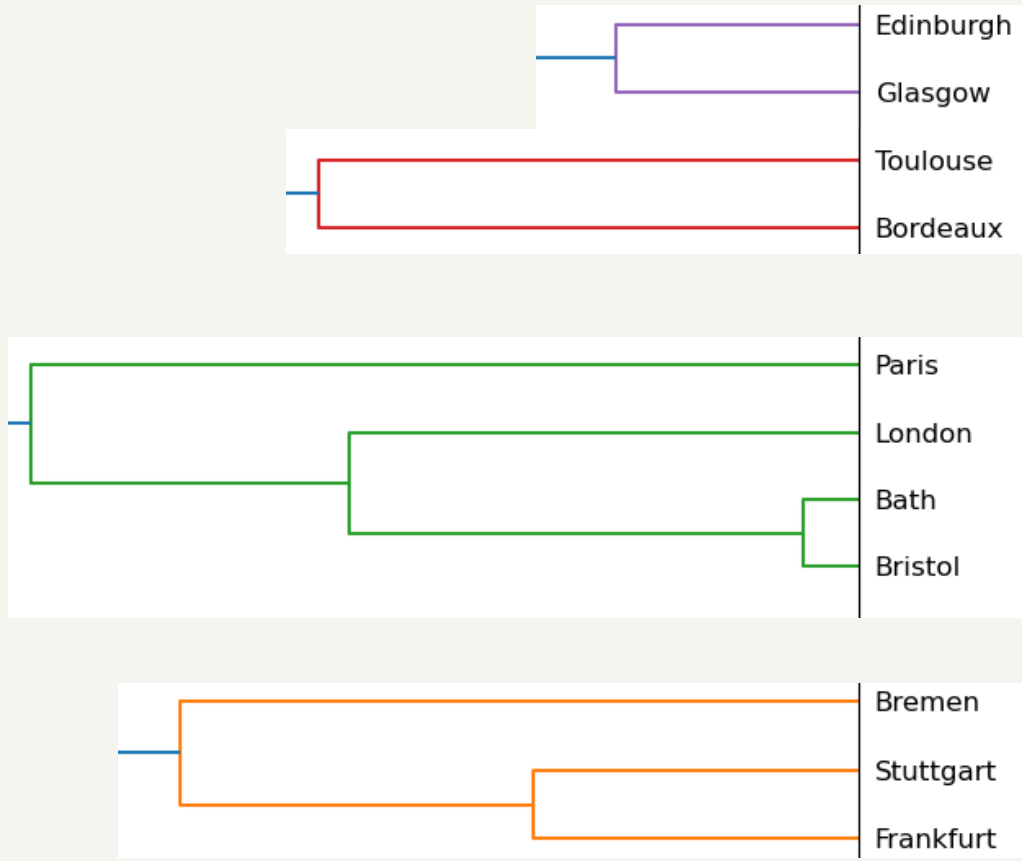
	[London, [Bristol, Bath]]	Paris	Lyon	Berlin	[[Frankfurt, Stuttgart], Bremen]	[Edinburgh, Glasgow]	[Toulouse, Bordeaux]
[London, [Bristol, Bath]]	0.0	3.6	7.6	13.6	8.9	4.5	6.7
Paris	3.6	0.0	4.0	11.6	6.5	9.0	5.0
Lyon	7.6	4.0	0.0	10.9	5.3	13.0	4.0
Berlin	13.6	11.6	10.9	0.0	4.6	16.9	14.9
[[Frankfurt, Stuttgart], Bremen]	8.9	6.5	5.3	4.6	0.0	12.3	9.3
[Edinburgh, Glasgow]	4.6	9.0	13.0	16.9	12.3	0.0	11.4
[Toulouse, Bordeaux]	6.7	5.0	4.0	14.9	9.3	11.4	0.0



Agglomerative Clustering

Merge the Clusters: Single Linkage

	[[[London , [Bristol, Bath]], Paris]	Lyon	Berlin	[[Frankfurt, Stuttgart], Bremen]	[Edinburgh, Glasgow]	[Toulouse, Bordeaux]
[[[London, Bristol, Bath]], Paris]	0.0	4.0	11.6	6.5	4.5	5.0
Lyon	4.0	0.0	10.9	5.3	13.0	4.0
Berlin	11.6	10.9	0.0	4.6	16.9	14.9
[[Frankfurt, Stuttgart], Bremen]	6.5	5.3	4.6	0.0	12.3	9.3
[Edinburgh, Glasgow]	4.6	13.0	16.9	12.3	0.0	11.4
[Toulouse, Bordeaux]	5.0	4.0	14.9	9.3	11.4	0.0



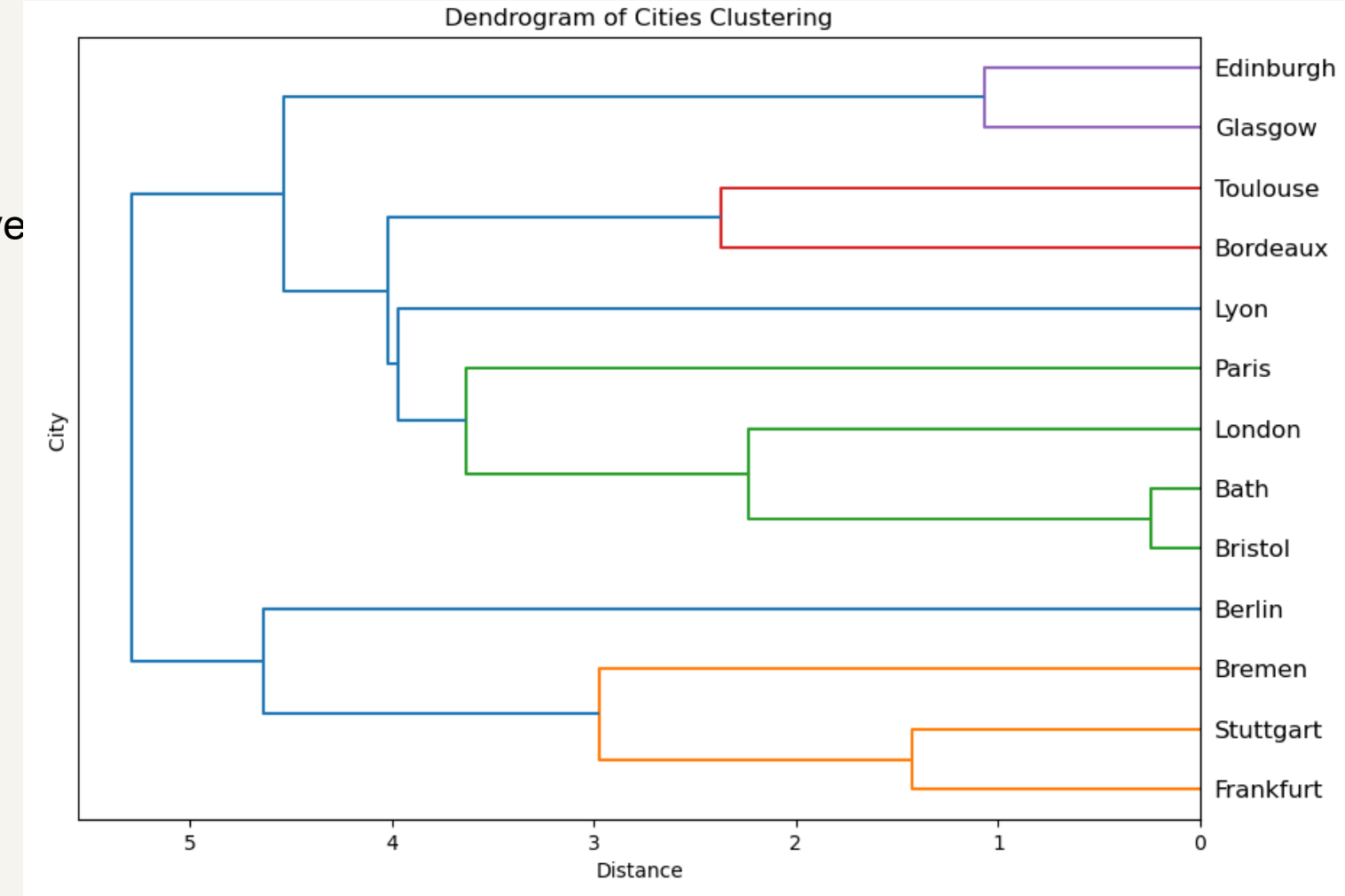
Agglomerative Clustering

Merge the Clusters: Single Linkage

We continue iterating the process until all rows and columns are merged and we have the final structure

Nested List Representation
(common in python implementations)

```
[[Berlin,
  [Bremen,
    [Stuttgart, Frankfurt],
  ],
 ],
 [[Edinburgh, Glasgow],
  [[Toulouse, Bordeaux],
   [Lyon,
     [Paris,
       [London,
         [Bristol, Bath]
       ]
     ]
   ]
  ]
 ]
]
```

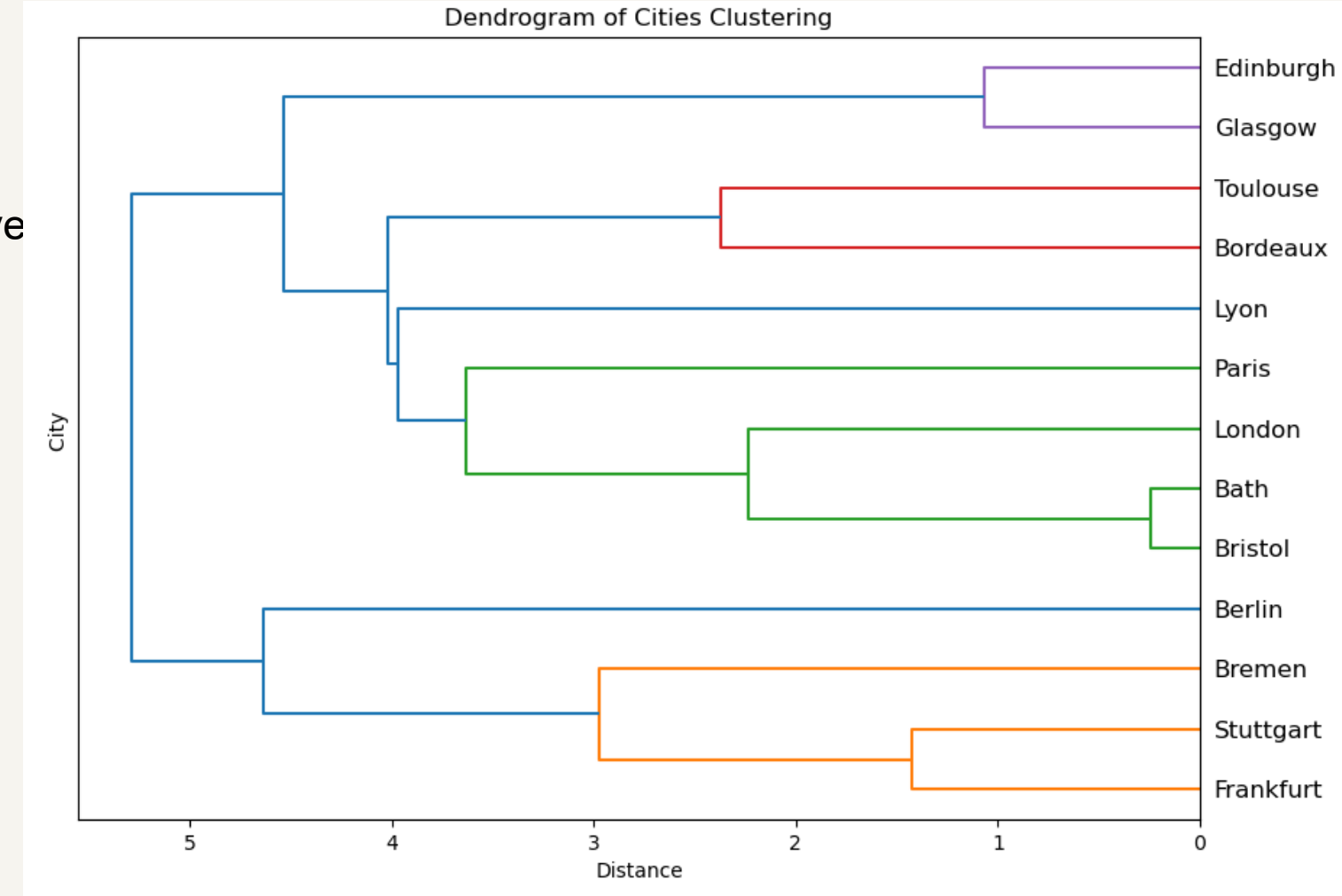


Agglomerative Clustering

Merge the Clusters: Single Linkage

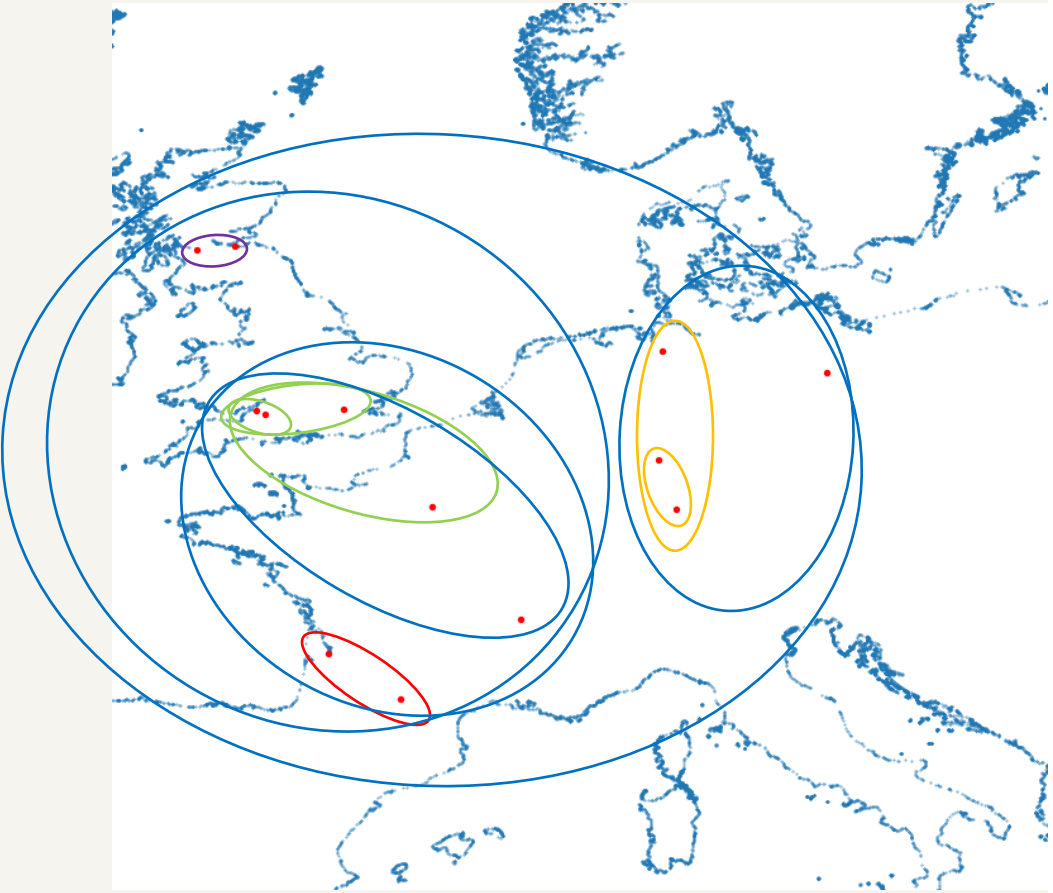
We continue iterating the process until all rows and columns are merged and we have the final structure

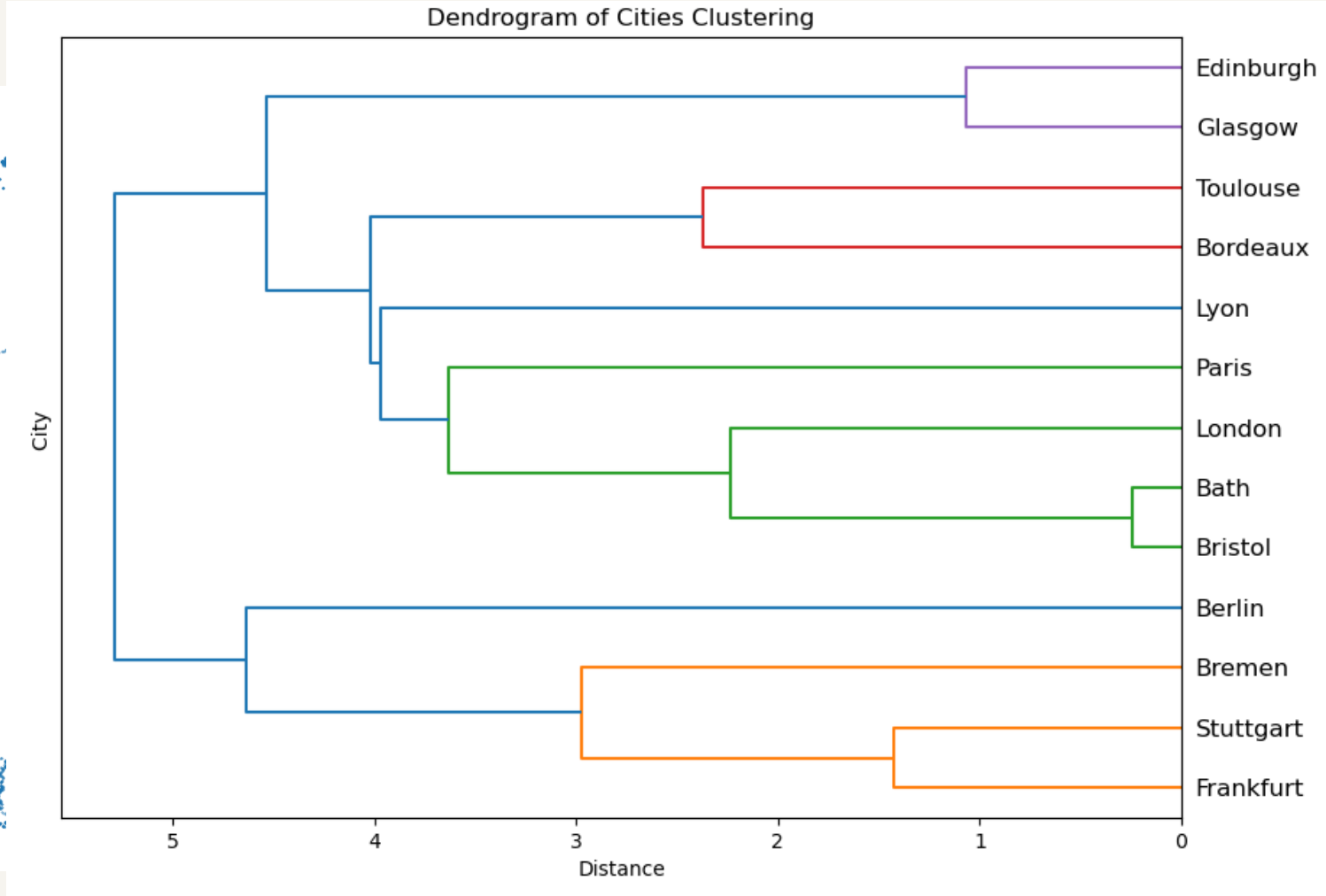
What do you think of the result?



Agglomerative Clustering

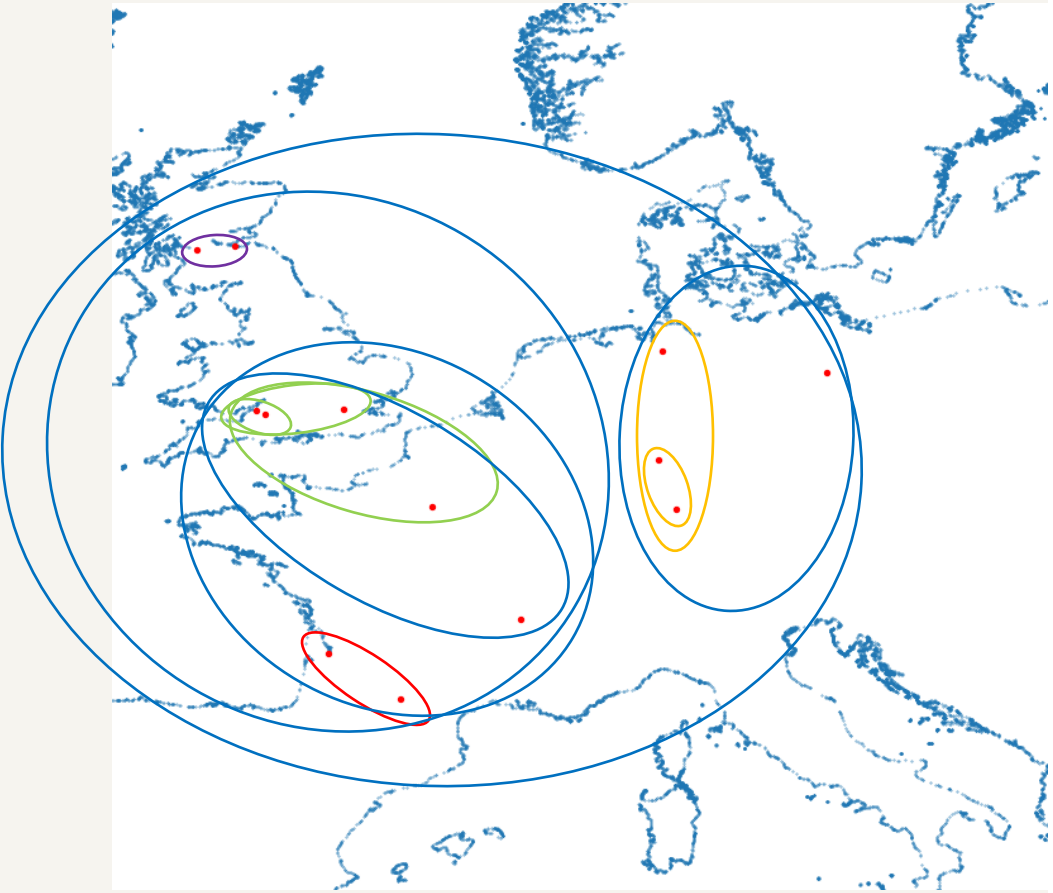
Merge the Clusters: Single Linkage

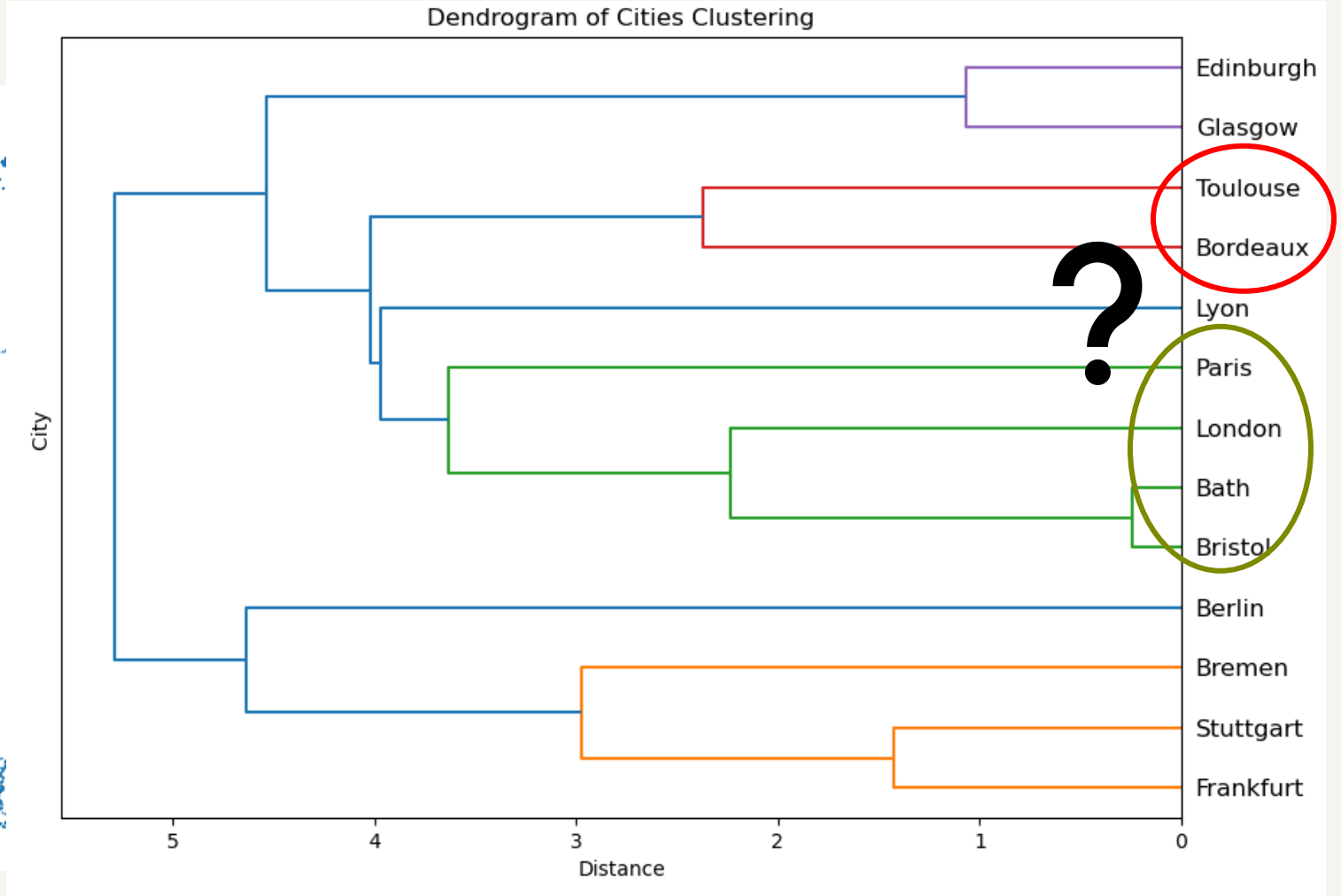




Agglomerative Clustering

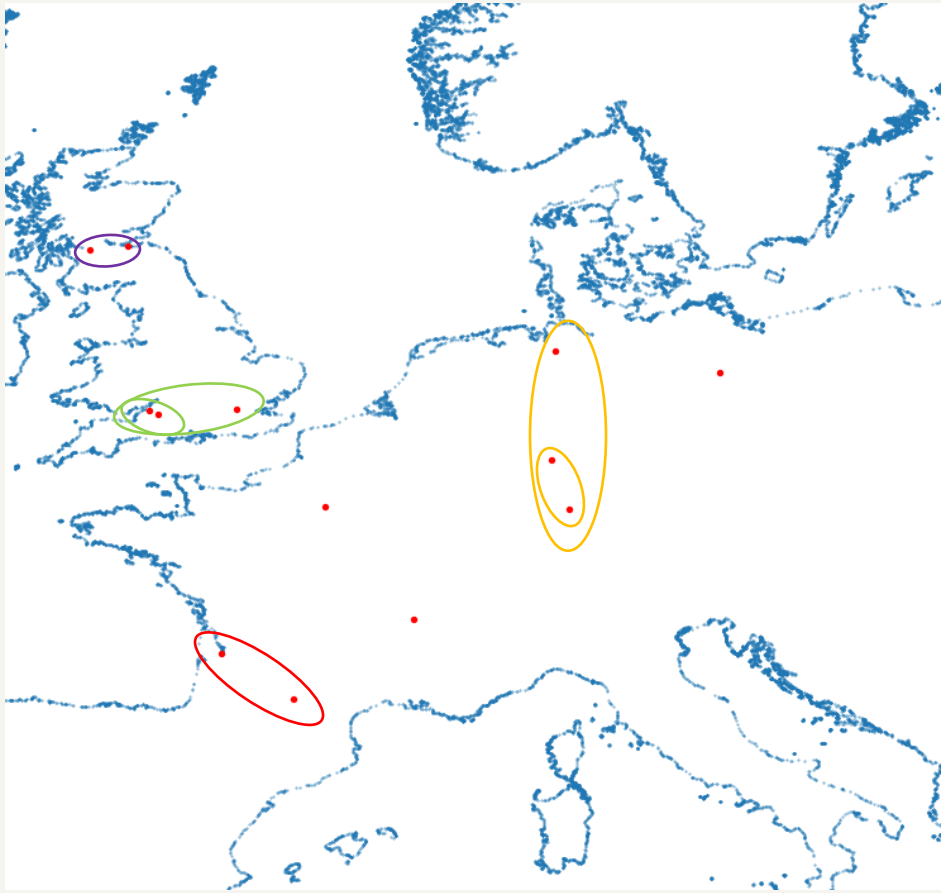
Merge the Clusters: Single Linkage

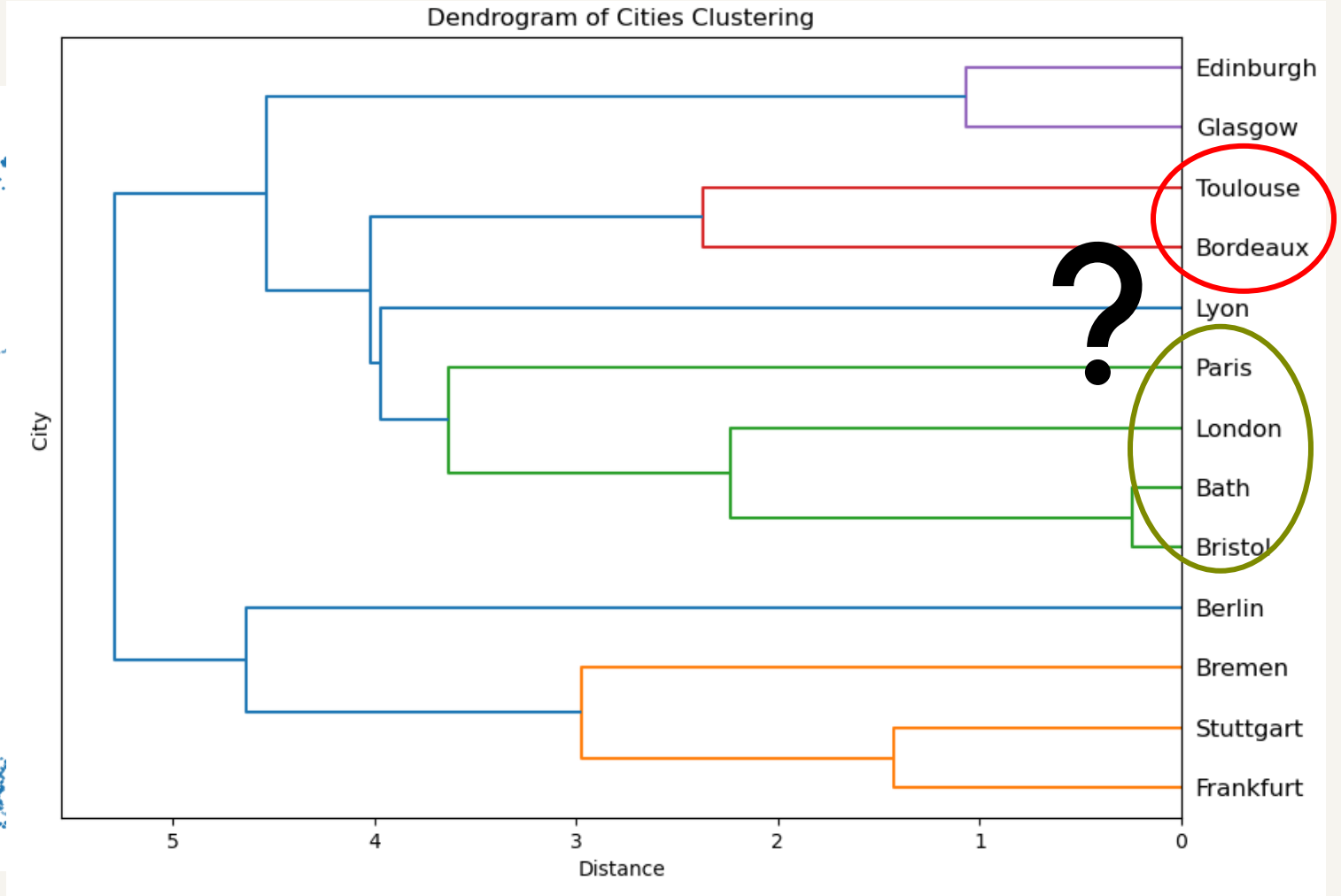




Agglomerative Clustering

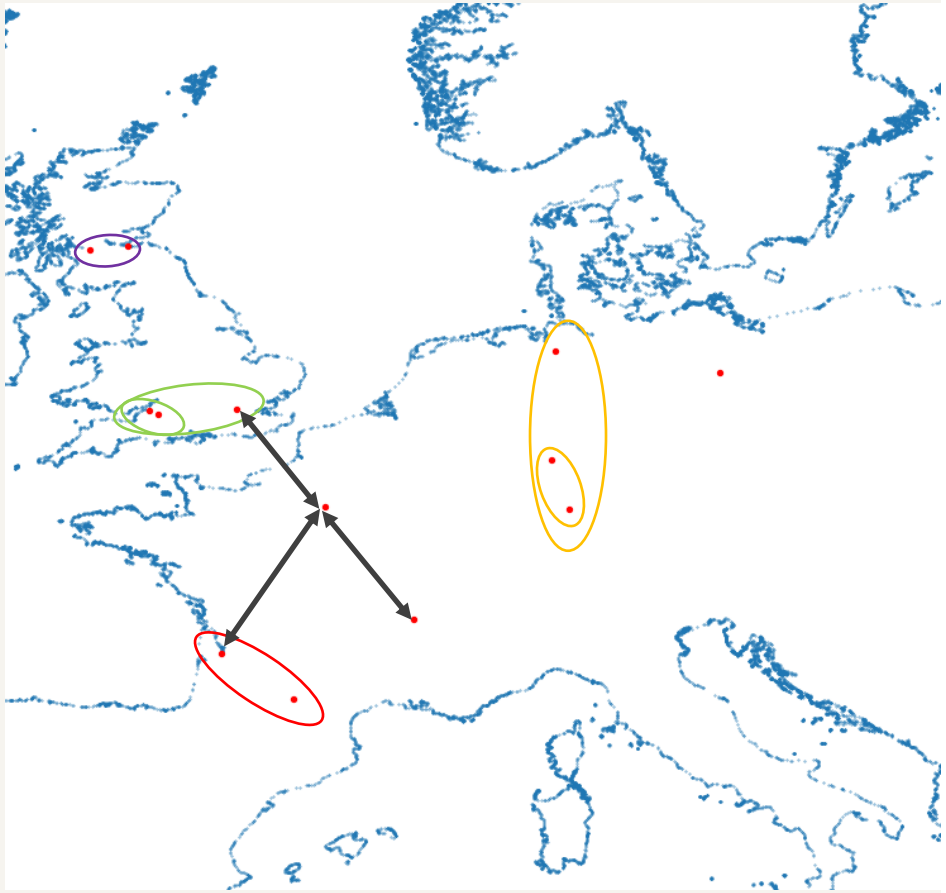
Merge the Clusters: Single Linkage

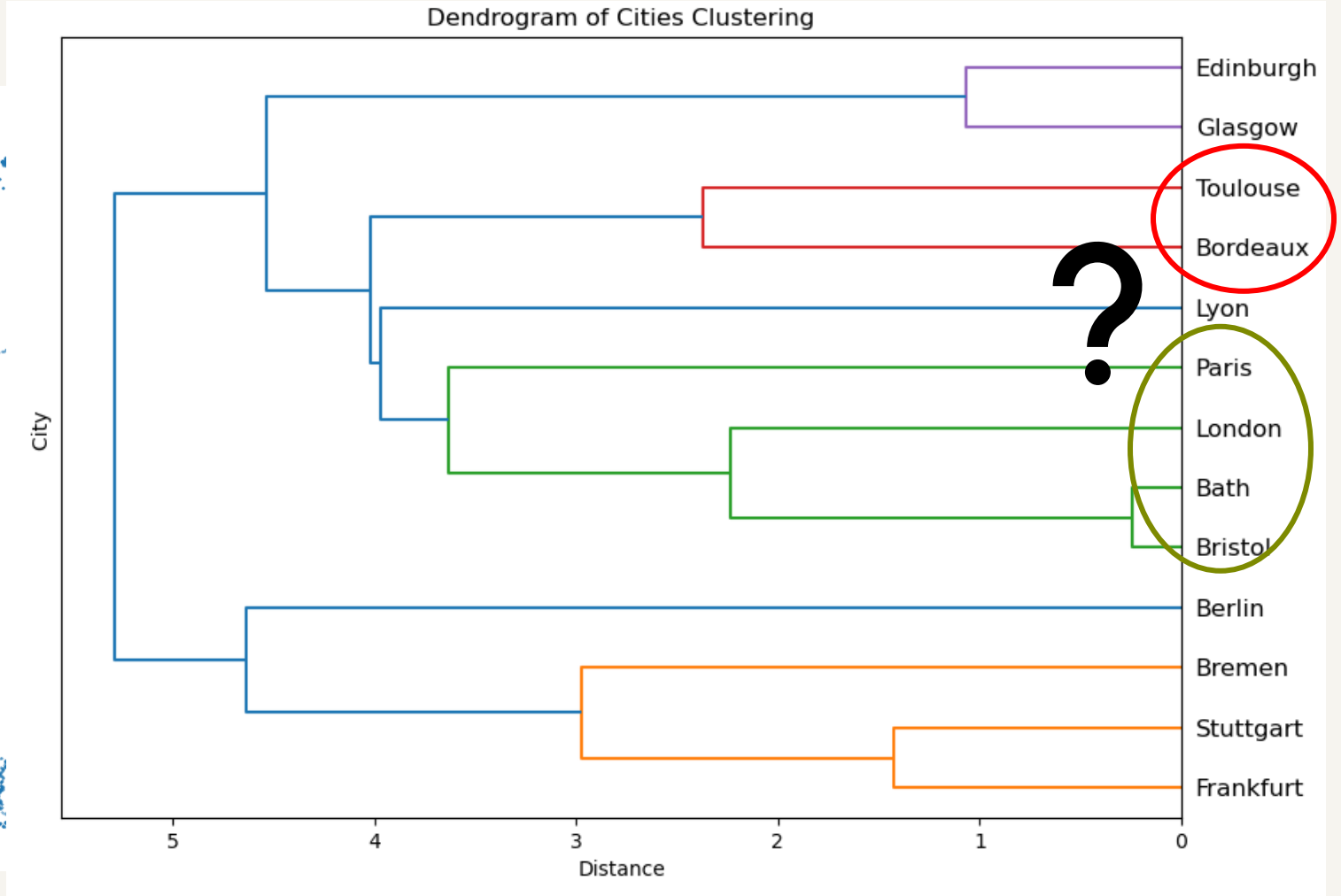




Agglomerative Clustering

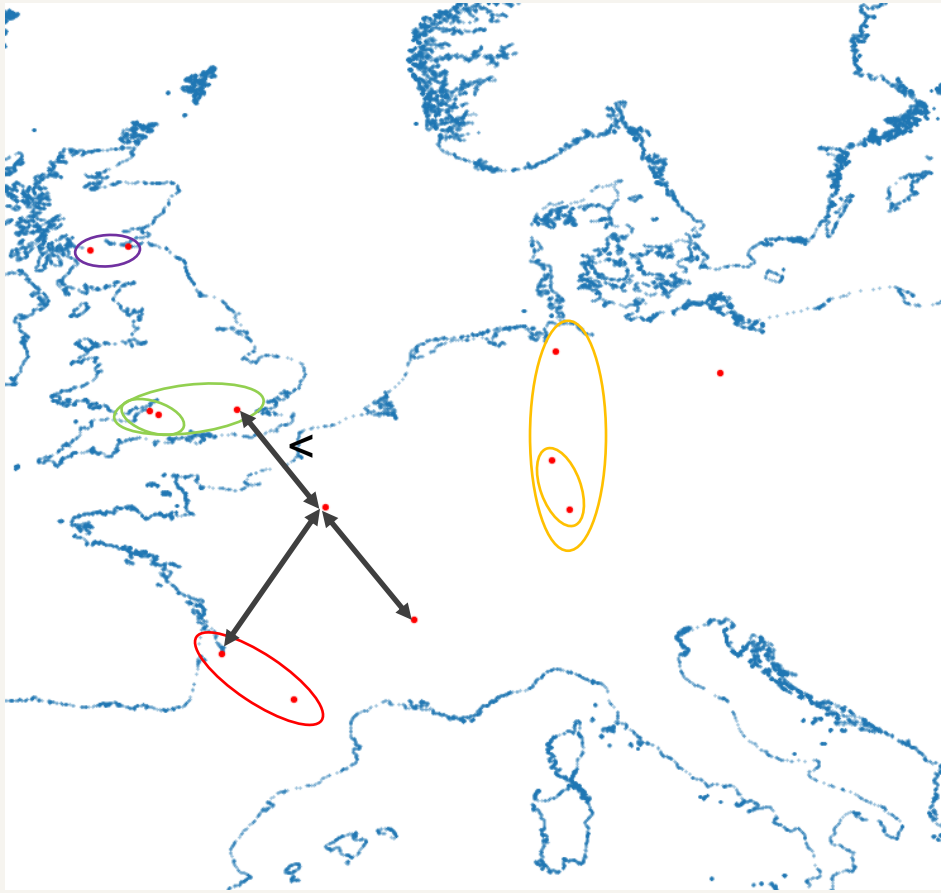
Merge the Clusters: Single Linkage

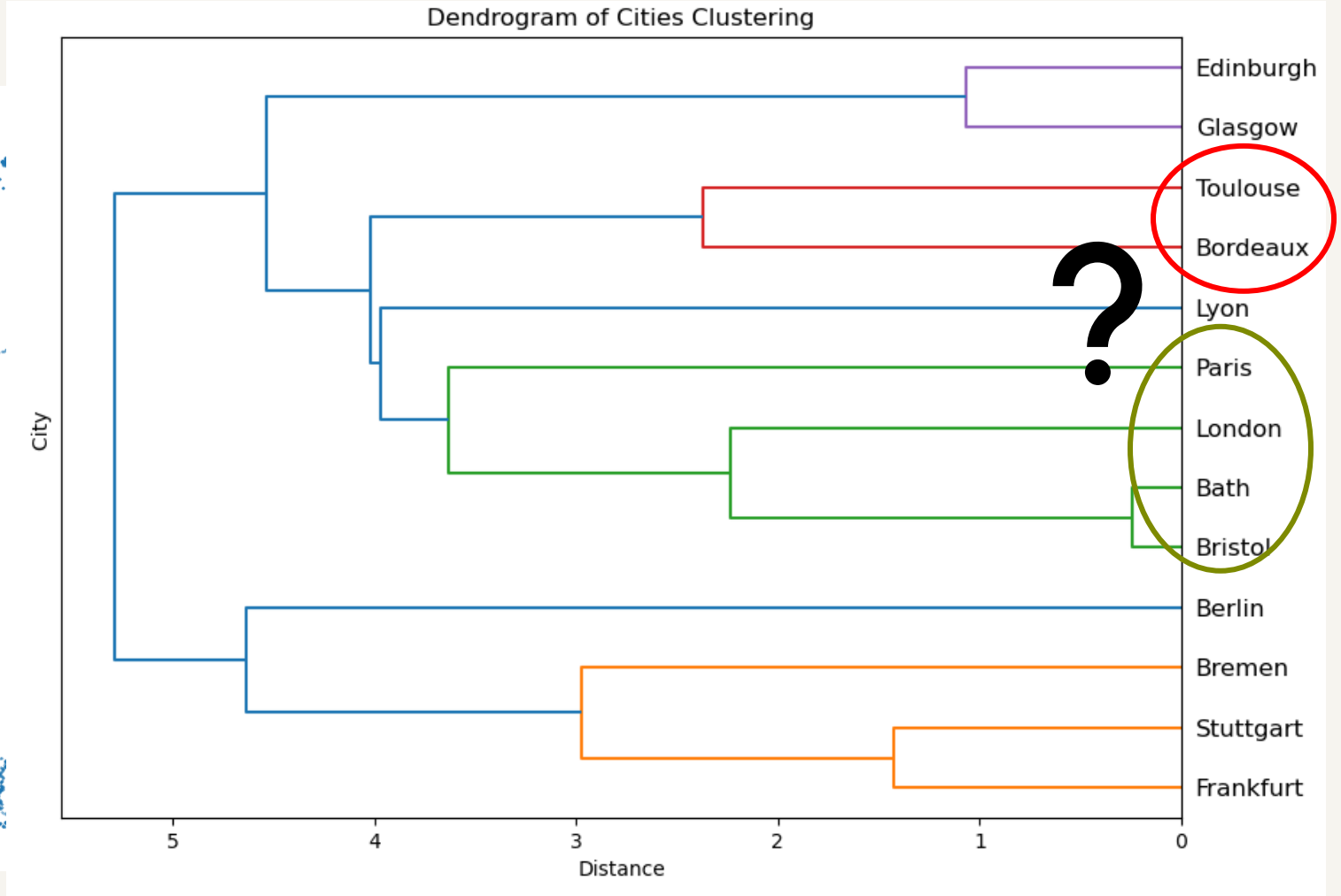




Agglomerative Clustering

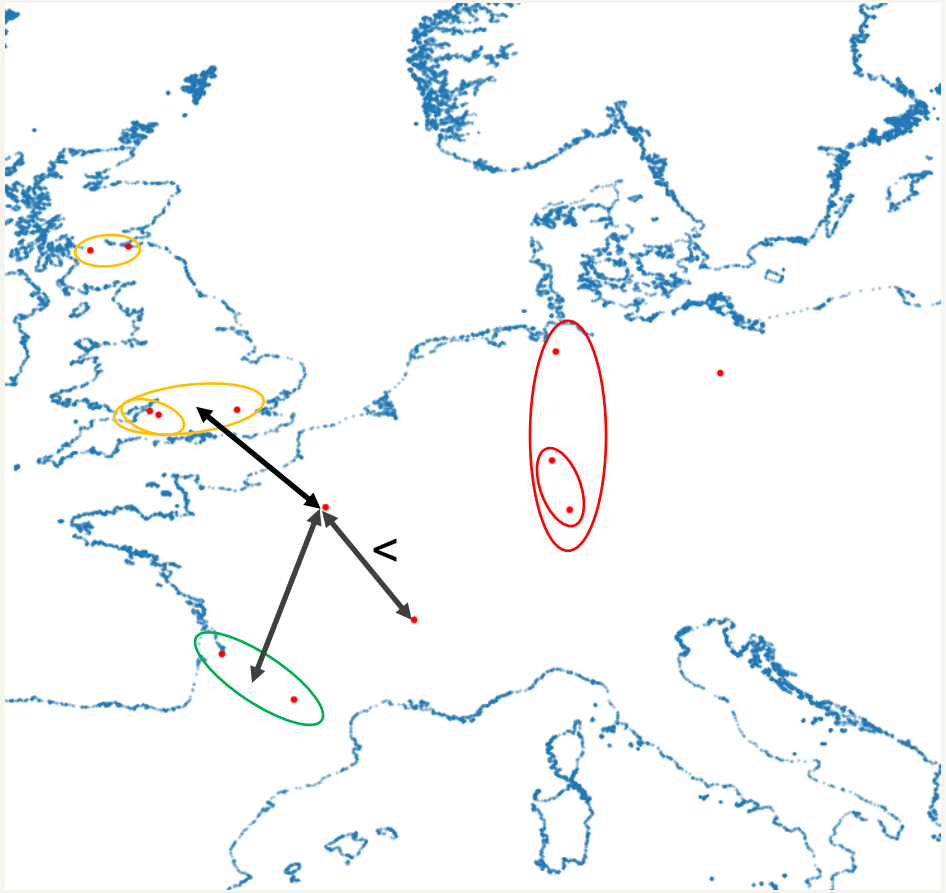
Merge the Clusters: Single Linkage

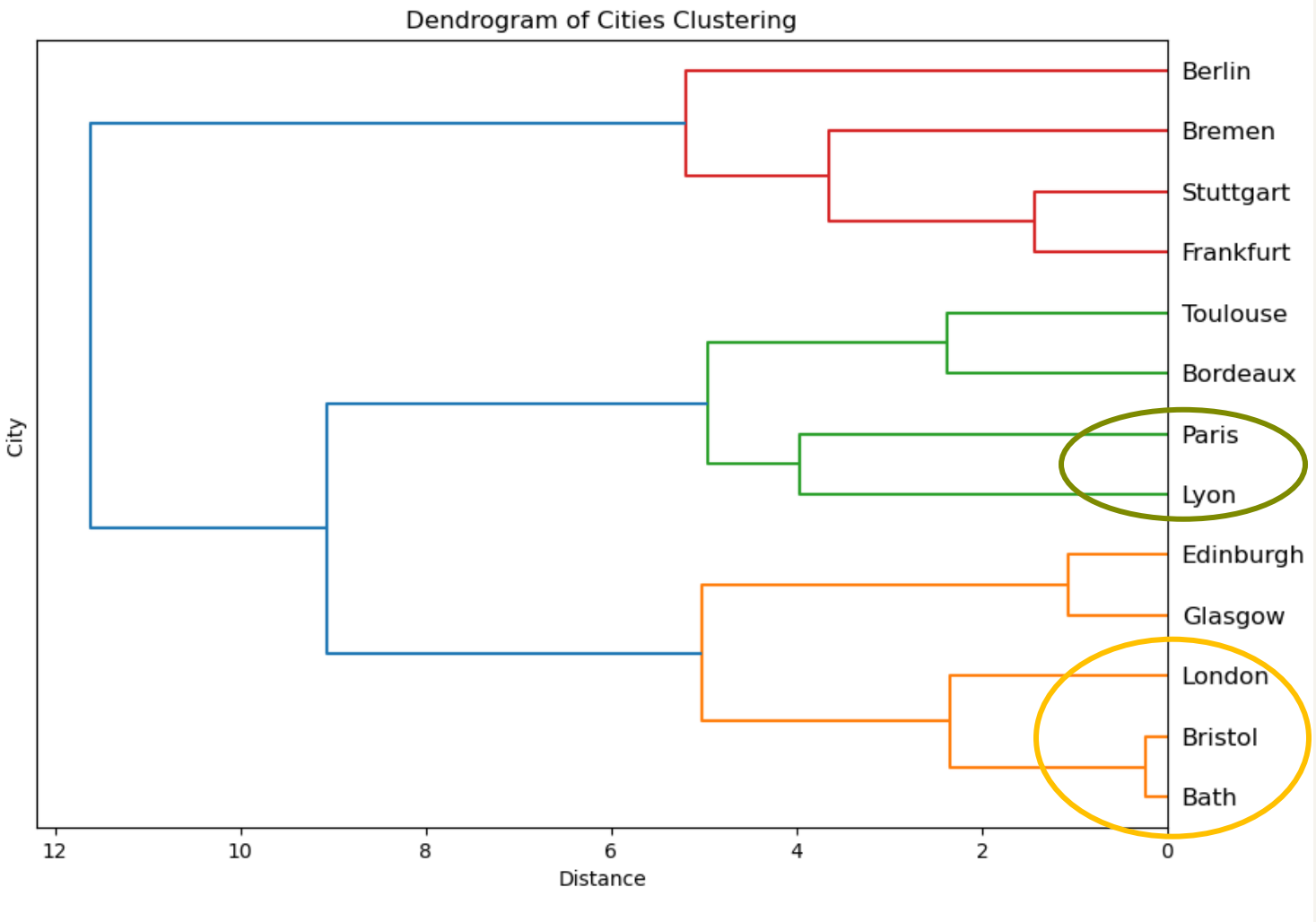




Agglomerative Clustering

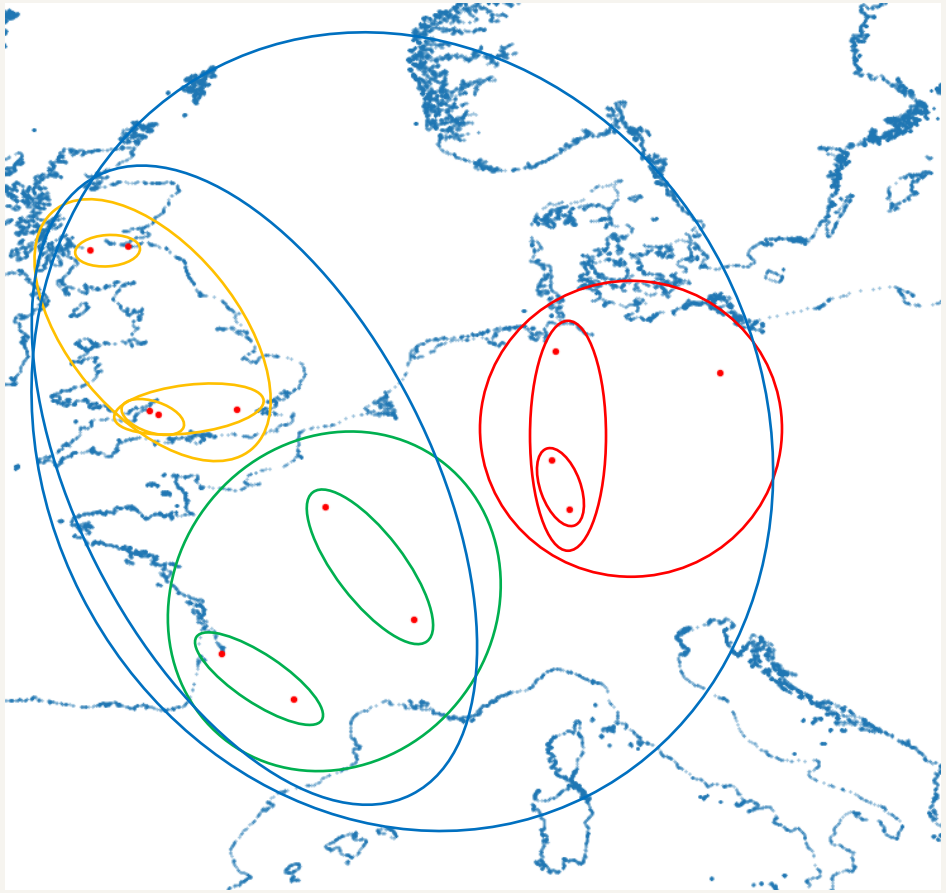
Merge the Clusters: Average Linkage

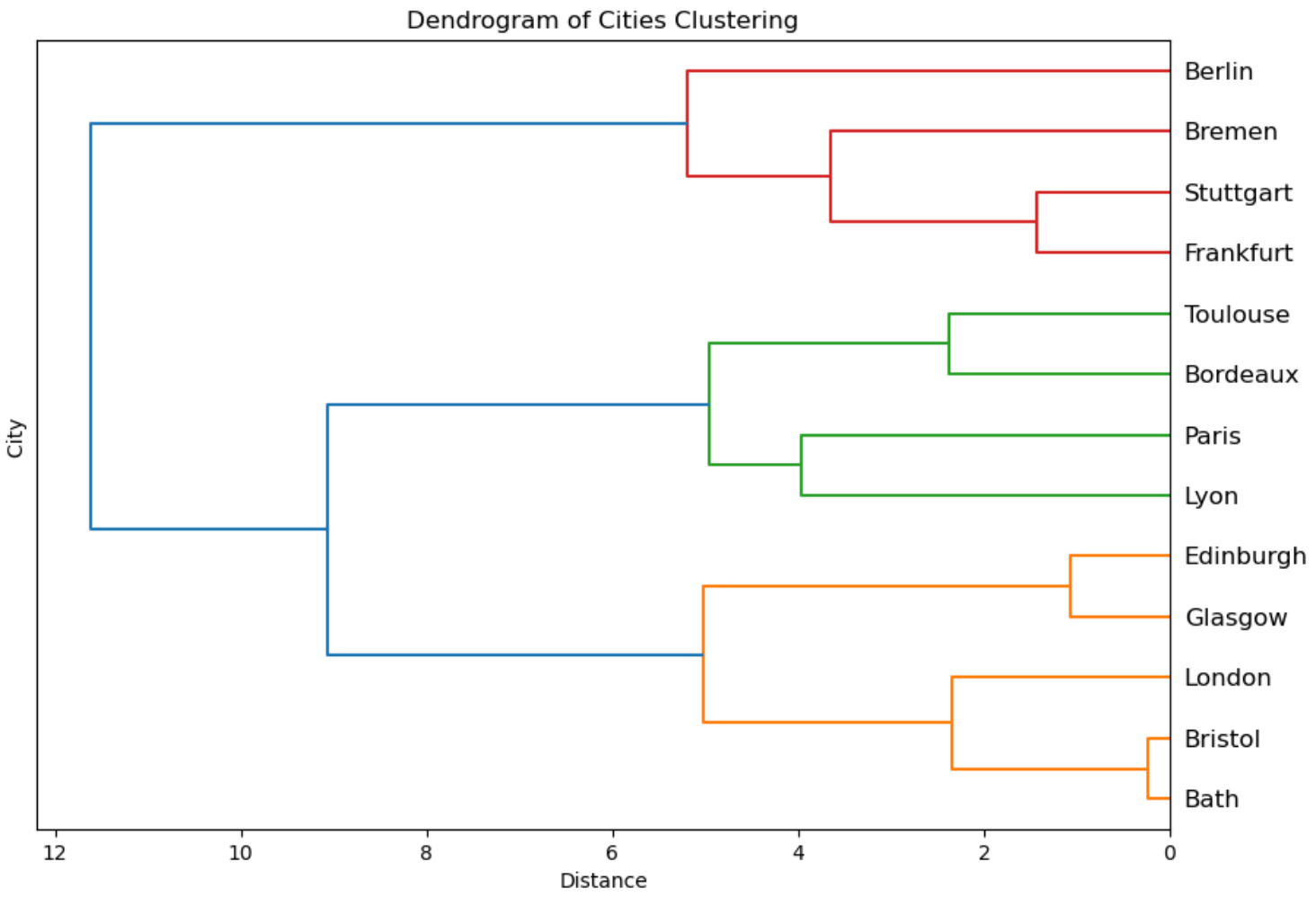




Agglomerative Clustering

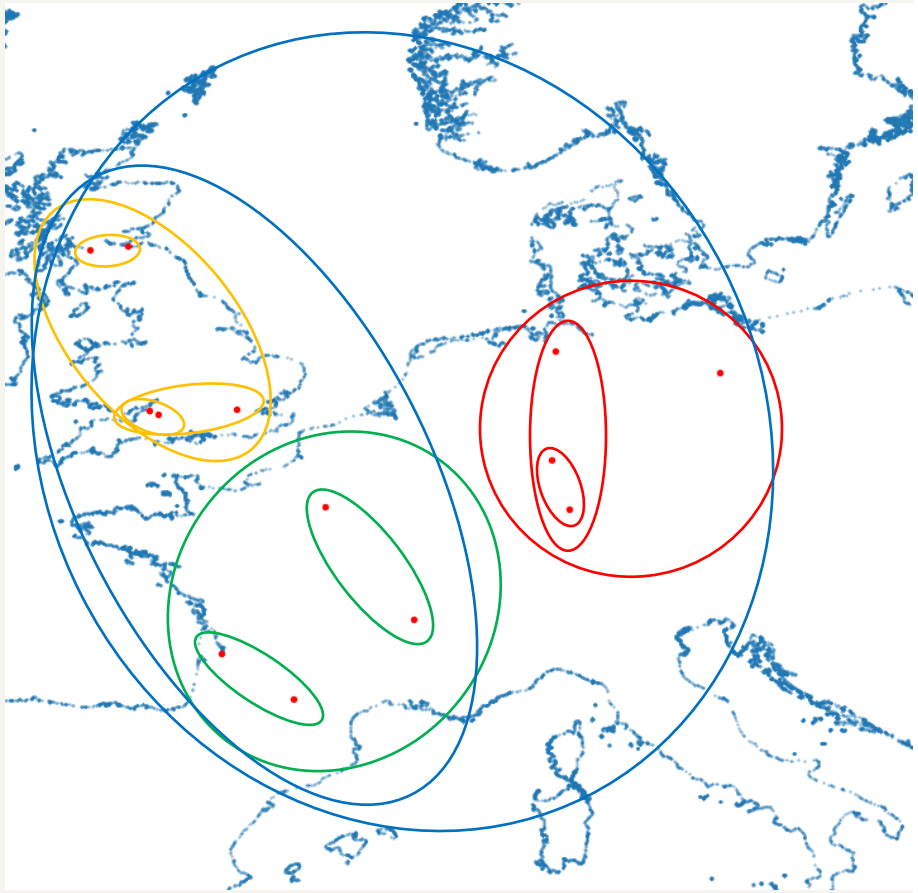
Merge the Clusters: Average Linkage

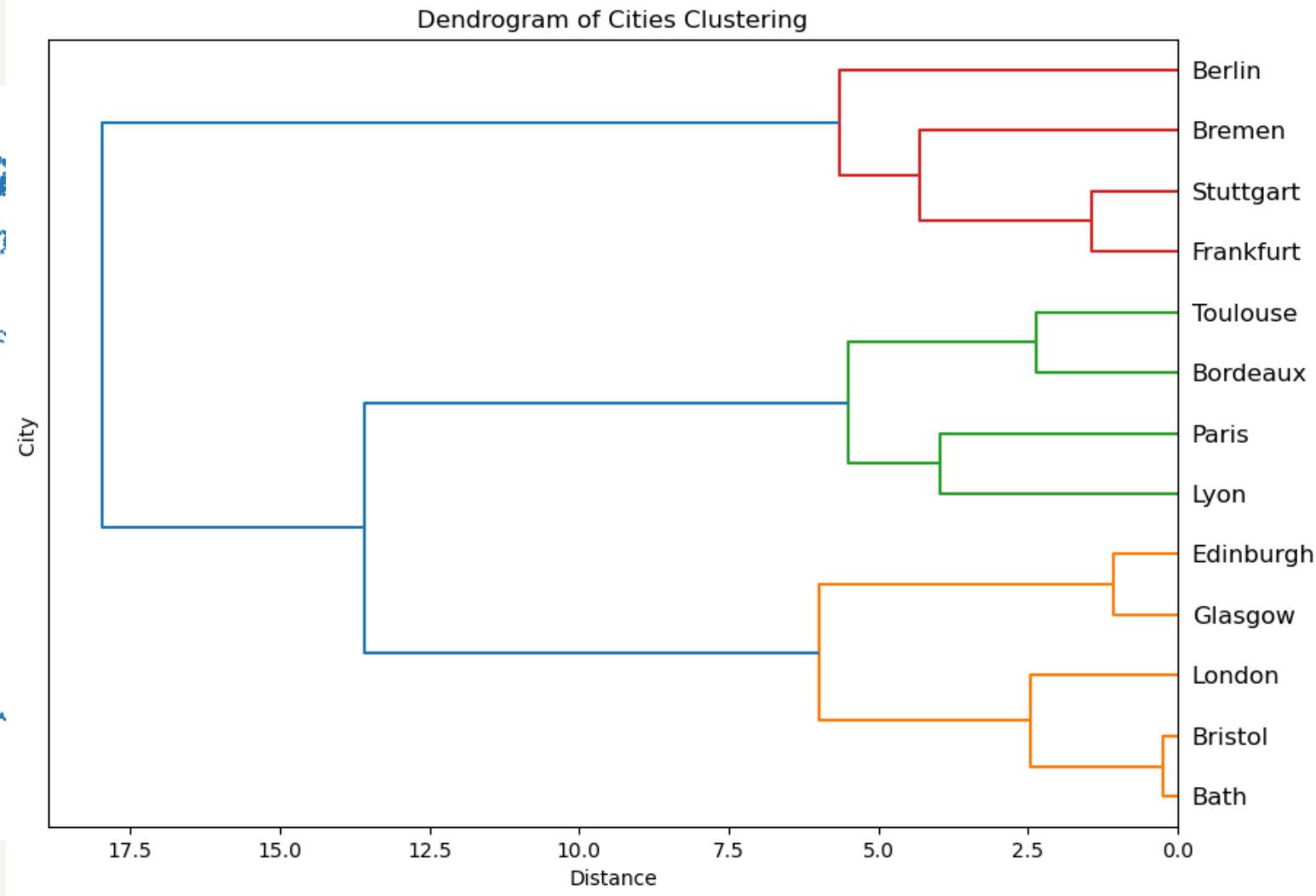




Agglomerative Clustering

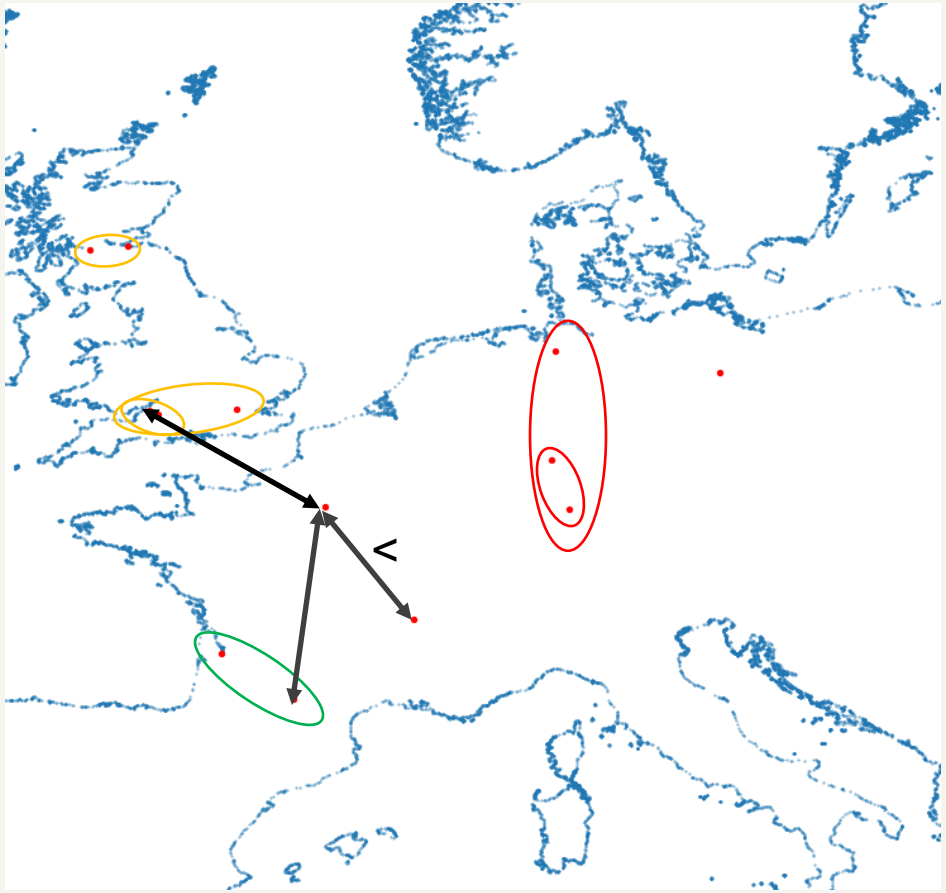
Merge the Clusters: Complete Linkage

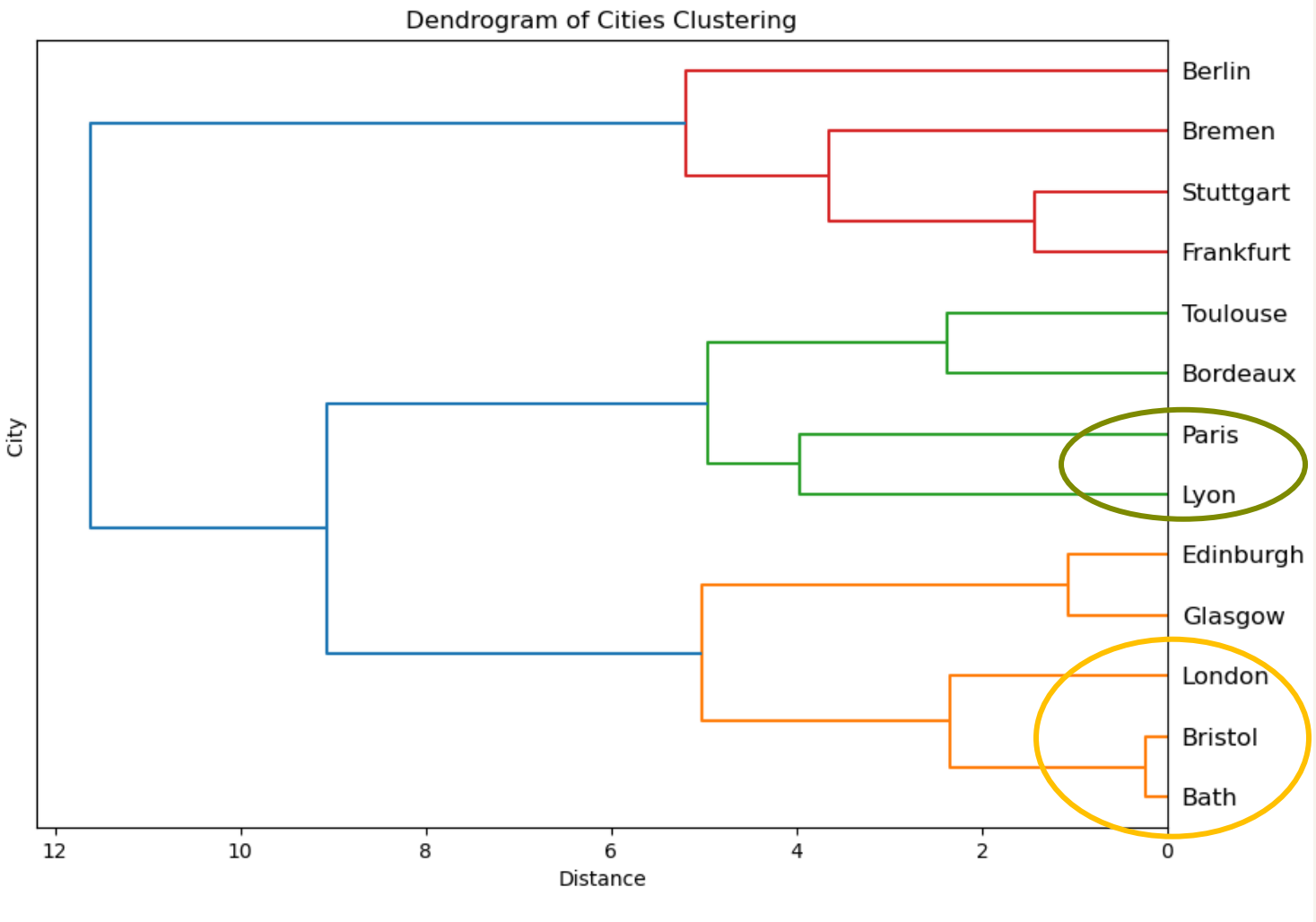




Agglomerative Clustering

Merge the Clusters: Average Linkage





Questions?

Divisive Clustering Example

Split data in two clusters:

- Find the most remote datapoint

	London	Bristol	Bath	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux
London	0.0	2.5	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7
Bristol	2.5	0.0	0.2	5.6	9.4	16.0	12.1	11.3	11.5	4.5	4.7	8.8	6.9
Bath	2.2	0.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.6	4.9	8.7	6.8
Paris	3.6	5.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0
Lyon	7.6	9.4	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5
Berlin	13.6	16.0	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0
Stuttgart	9.7	12.1	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5
Frankfurt	8.9	11.3	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7
Bremen	9.1	11.5	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5
Edinburgh	5.4	4.5	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4
Glasgow	6.0	4.7	4.9	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6
Toulouse	8.1	8.8	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4
Bordeaux	6.7	6.9	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0

Divisive Clustering Example

Split data in two clusters:

- Find the most remote datapoint
- Average over columns

	London	Bristol	Bath	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux	AVG
London	0.0	2.5	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7	**6.4**
Bristol	2.5	0.0	0.2	5.6	9.4	16.0	12.1	11.3	11.5	4.5	4.7	8.8	6.9	**7.2**
Bath	2.2	0.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.6	4.9	8.7	6.8	**7.1**
Paris	3.6	5.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0	**6.2**
Lyon	7.6	9.4	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5	**7.4**
Berlin	13.6	16.0	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0	**11.5**
Stuttgart	9.7	12.1	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5	**8.2**
Frankfurt	8.9	11.3	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7	**7.8**
Bremen	9.1	11.5	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5	**8.5**
Edinburgh	5.4	4.5	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4	**9.2**
Glasgow	6.0	4.7	4.9	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6	**9.7**
Toulouse	8.1	8.8	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4	**8.5**
Bordeaux	6.7	6.9	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0	**8.1**

Divisive Clustering Example

Split data in two clusters:

- Find the most remote datapoint
- Average over columns -> select the greatest

	London	Bristol	Bath	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux	AVG
London	0.0	2.5	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7	**6.4**
Bristol	2.5	0.0	0.2	5.6	9.4	16.0	12.1	11.3	11.5	4.5	4.7	8.8	6.9	**7.2**
Bath	2.2	0.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.6	4.9	8.7	6.8	**7.1**
Paris	3.6	5.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0	**6.2**
Lyon	7.6	9.4	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5	**7.4**
Berlin	13.6	16.0	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0	**11.5**
Stuttgart	9.7	12.1	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5	**8.2**
Frankfurt	8.9	11.3	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7	**7.8**
Bremen	9.1	11.5	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5	**8.5**
Edinburgh	5.4	4.5	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4	**9.2**
Glasgow	6.0	4.7	4.9	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6	**9.7**
Toulouse	8.1	8.8	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4	**8.5**
Bordeaux	6.7	6.9	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0	**8.1**

Divisive Clustering Example

Split data in two clusters:

- Find the most remote datapoint
- Average over columns -> select the greatest and use it to define a new cluster

	London	Bristol	Bath	Paris	Lyon	Berlin	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux	AVG
London	0.0	2.5	2.2	3.6	7.6	13.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7	**6.4**
Bristol	2.5	0.0	0.2	5.6	9.4	16.0	12.1	11.3	11.5	4.5	4.7	8.8	6.9	**7.2**
Bath	2.2	0.2	0.0	5.3	9.1	15.8	11.8	11.1	11.3	4.6	4.9	8.7	6.8	**7.1**
Paris	3.6	5.6	5.3	0.0	4.0	11.6	6.8	6.5	7.7	9.0	9.6	5.3	5.0	**6.2**
Lyon	7.6	9.4	9.1	4.0	0.0	10.9	5.3	5.8	8.3	13.0	13.6	4.0	5.5	**7.4**
Berlin	13.6	16.0	15.8	11.6	10.9	0.0	5.6	5.3	4.6	16.9	18.0	14.9	16.0	**11.5**
Stuttgart	9.7	12.1	11.8	6.8	5.3	5.6	0.0	1.4	4.3	14.3	15.2	9.3	10.5	**8.2**
Frankfurt	8.9	11.3	11.1	6.5	5.8	5.3	1.4	0.0	3.0	13.2	14.2	9.7	10.7	**7.8**
Bremen	9.1	11.5	11.3	7.7	8.3	4.6	4.3	3.0	0.0	12.3	13.3	12.0	12.5	**8.5**
Edinburgh	5.4	4.5	4.6	9.0	13.0	16.9	14.3	13.2	12.3	0.0	1.1	13.2	11.4	**9.2**
Glasgow	6.0	4.7	4.9	9.6	13.6	18.0	15.2	14.2	13.3	1.1	0.0	13.5	11.6	**9.7**
Toulouse	8.1	8.8	8.7	5.3	4.0	14.9	9.3	9.7	12.0	13.2	13.5	0.0	2.4	**8.5**
Bordeaux	6.7	6.9	6.8	5.0	5.5	16.0	10.5	10.7	12.5	11.4	11.6	2.4	0.0	**8.1**

Divisive Clustering Example

Split data in two clusters:

- Find the most remote datapoint
- Average over columns -> select the greatest and use it to define a new cluster

	London	Bristol	Bath	Paris	Lyon	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux	AVG	Berlin
London	0.0	2.5	2.2	3.6	7.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7	**5.8**	13.6
Bristol	2.5	0.0	0.2	5.6	9.4	12.1	11.3	11.5	4.5	4.7	8.8	6.9	**6.5**	16.0
Bath	2.2	0.2	0.0	5.3	9.1	11.8	11.1	11.3	4.6	4.9	8.7	6.8	**6.3**	15.8
Paris	3.6	5.6	5.3	0.0	4.0	6.8	6.5	7.7	9.0	9.6	5.3	5.0	**5.7**	11.6
Lyon	7.6	9.4	9.1	4.0	0.0	5.3	5.8	8.3	13.0	13.6	4.0	5.5	**7.1**	10.9
Stuttgart	9.7	12.1	11.8	6.8	5.3	0.0	1.4	4.3	14.3	15.2	9.3	10.5	**8.4**	5.6
Frankfurt	8.9	11.3	11.1	6.5	5.8	1.4	0.0	3.0	13.2	14.2	9.7	10.7	**8.0**	5.3
Bremen	9.1	11.5	11.3	7.7	8.3	4.3	3.0	0.0	12.3	13.3	12.0	12.5	**8.8**	4.6
Edinburgh	5.4	4.5	4.6	9.0	13.0	14.3	13.2	12.3	0.0	1.1	13.2	11.4	**8.5**	16.9
Glasgow	6.0	4.7	4.9	9.6	13.6	15.2	14.2	13.3	1.1	0.0	13.5	11.6	**9.0**	18.0
Toulouse	8.1	8.8	8.7	5.3	4.0	9.3	9.7	12.0	13.2	13.5	0.0	2.4	**7.9**	14.9
Bordeaux	6.7	6.9	6.8	5.0	5.5	10.5	10.7	12.5	11.4	11.6	2.4	0.0	**7.5**	16.0

Divisive Clustering Example

Split data in two clusters:

- Compare the distances of all datapoints to the center of each of the two clusters

	London	Bristol	Bath	Paris	Lyon	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux	AVG	Berlin
London	0.0	2.5	2.2	3.6	7.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7	**5.8**	13.6
Bristol	2.5	0.0	0.2	5.6	9.4	12.1	11.3	11.5	4.5	4.7	8.8	6.9	**6.5**	16.0
Bath	2.2	0.2	0.0	5.3	9.1	11.8	11.1	11.3	4.6	4.9	8.7	6.8	**6.3**	15.8
Paris	3.6	5.6	5.3	0.0	4.0	6.8	6.5	7.7	9.0	9.6	5.3	5.0	**5.7**	11.6
Lyon	7.6	9.4	9.1	4.0	0.0	5.3	5.8	8.3	13.0	13.6	4.0	5.5	**7.1**	10.9
Stuttgart	9.7	12.1	11.8	6.8	5.3	0.0	1.4	4.3	14.3	15.2	9.3	10.5	**8.4**	5.6
Frankfurt	8.9	11.3	11.1	6.5	5.8	1.4	0.0	3.0	13.2	14.2	9.7	10.7	**8.0**	5.3
Bremen	9.1	11.5	11.3	7.7	8.3	4.3	3.0	0.0	12.3	13.3	12.0	12.5	**8.8**	4.6
Edinburgh	5.4	4.5	4.6	9.0	13.0	14.3	13.2	12.3	0.0	1.1	13.2	11.4	**8.5**	16.9
Glasgow	6.0	4.7	4.9	9.6	13.6	15.2	14.2	13.3	1.1	0.0	13.5	11.6	**9.0**	18.0
Toulouse	8.1	8.8	8.7	5.3	4.0	9.3	9.7	12.0	13.2	13.5	0.0	2.4	**7.9**	14.9
Bordeaux	6.7	6.9	6.8	5.0	5.5	10.5	10.7	12.5	11.4	11.6	2.4	0.0	**7.5**	16.0

Divisive Clustering Example

Split data in two clusters:

- Compare the distances of all datapoints to the center of each of the two clusters
- Reassign the ones that are closer to the new cluster

	London	Bristol	Bath	Paris	Lyon	Stuttgart	Frankfurt	Bremen	Edinburgh	Glasgow	Toulouse	Bordeaux	AVG	Berlin
London	0.0	2.5	2.2	3.6	7.6	9.7	8.9	9.1	5.4	6.0	8.1	6.7	**5.8**	13.6
Bristol	2.5	0.0	0.2	5.6	9.4	12.1	11.3	11.5	4.5	4.7	8.8	6.9	**6.5**	16.0
Bath	2.2	0.2	0.0	5.3	9.1	11.8	11.1	11.3	4.6	4.9	8.7	6.8	**6.3**	15.8
Paris	3.6	5.6	5.3	0.0	4.0	6.8	6.5	7.7	9.0	9.6	5.3	5.0	**5.7**	11.6
Lyon	7.6	9.4	9.1	4.0	0.0	5.3	5.8	8.3	13.0	13.6	4.0	5.5	**7.1**	10.9
Stuttgart	9.7	12.1	11.8	6.8	5.3	0.0	1.4	4.3	14.3	15.2	9.3	10.5	**8.4**	5.6
Frankfurt	8.9	11.3	11.1	6.5	5.8	1.4	0.0	3.0	13.2	14.2	9.7	10.7	**8.0**	5.3
Bremen	9.1	11.5	11.3	7.7	8.3	4.3	3.0	0.0	12.3	13.3	12.0	12.5	**8.8**	4.6
Edinburgh	5.4	4.5	4.6	9.0	13.0	14.3	13.2	12.3	0.0	1.1	13.2	11.4	**8.5**	16.9
Glasgow	6.0	4.7	4.9	9.6	13.6	15.2	14.2	13.3	1.1	0.0	13.5	11.6	**9.0**	18.0
Toulouse	8.1	8.8	8.7	5.3	4.0	9.3	9.7	12.0	13.2	13.5	0.0	2.4	**7.9**	14.9
Bordeaux	6.7	6.9	6.8	5.0	5.5	10.5	10.7	12.5	11.4	11.6	2.4	0.0	**7.5**	16.0

Divisive Clustering Example

Split data in two clusters:

- Compare the distances of all datapoints to the center of each of the two clusters
- Reassign the ones that are closer to the new cluster

	London	Bristol	Bath	Paris	Lyon	Edinburgh	Glasgow	Toulouse	Bordeaux	AVG
London	0.0	2.5	2.2	3.6	7.6	5.4	6.0	8.1	6.7	5.8
Bristol	2.5	0.0	0.2	5.6	9.4	4.5	4.7	8.8	6.9	6.5
Bath	2.2	0.2	0.0	5.3	9.1	4.6	4.9	8.7	6.8	6.3
Paris	3.6	5.6	5.3	0.0	4.0	9.0	9.6	5.3	5.0	5.7
Lyon	7.6	9.4	9.1	4.0	0.0	13.0	13.6	4.0	5.5	7.1
Edinburgh	5.4	4.5	4.6	9.0	13.0	0.0	1.1	13.2	11.4	8.5
Glasgow	6.0	4.7	4.9	9.6	13.6	1.1	0.0	13.5	11.6	9.0
Toulouse	8.1	8.8	8.7	5.3	4.0	13.2	13.5	0.0	2.4	7.9
Bordeaux	6.7	6.9	6.8	5.0	5.5	11.4	11.6	2.4	0.0	7.5

	Berlin	Stuttgart	Frankfurt	Bremen	AVG
Berlin	0.0	5.6	5.3	4.6	3.9
Stuttgart	5.6	0.0	1.4	4.3	2.8
Frankfurt	5.3	1.4	0.0	3.0	2.4
Bremen	4.6	4.3	3.0	0.0	3

Divisive Clustering Example

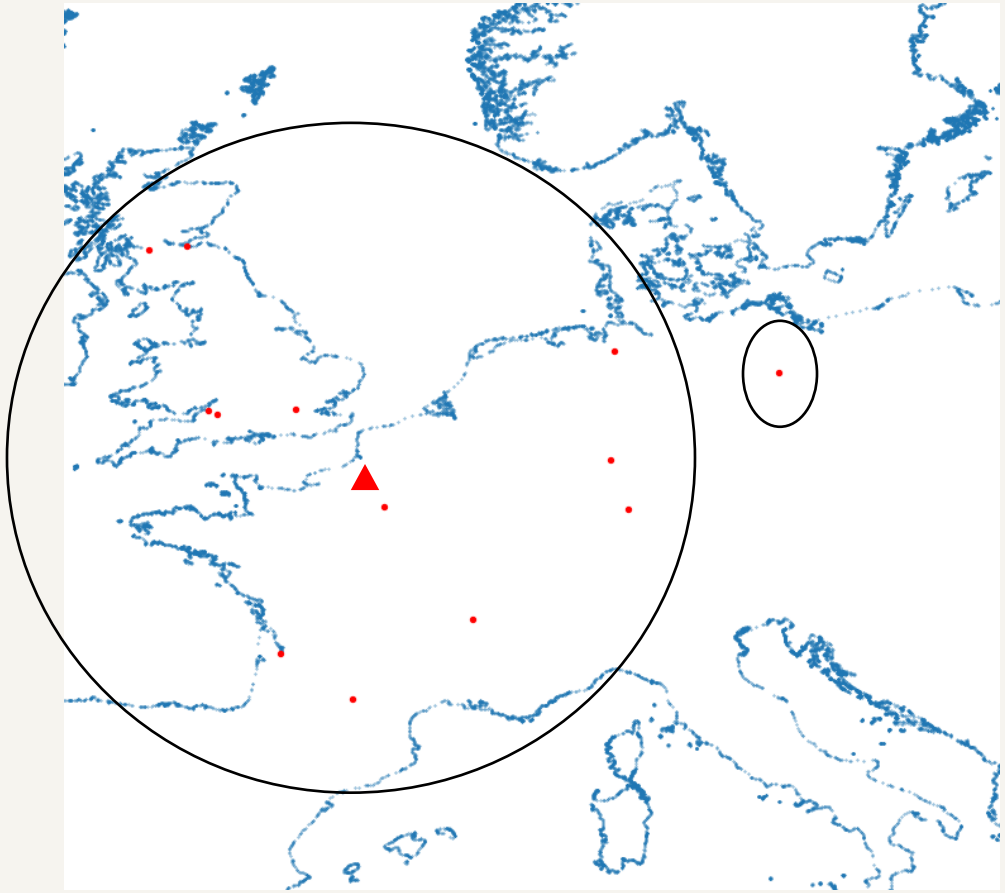
- Calculate the distance of the datapoints to the centre of the cluster

Divisive Clustering Example

- Calculate the distance of the datapoints to the centre of the cluster
- Identify the most remote datapoint

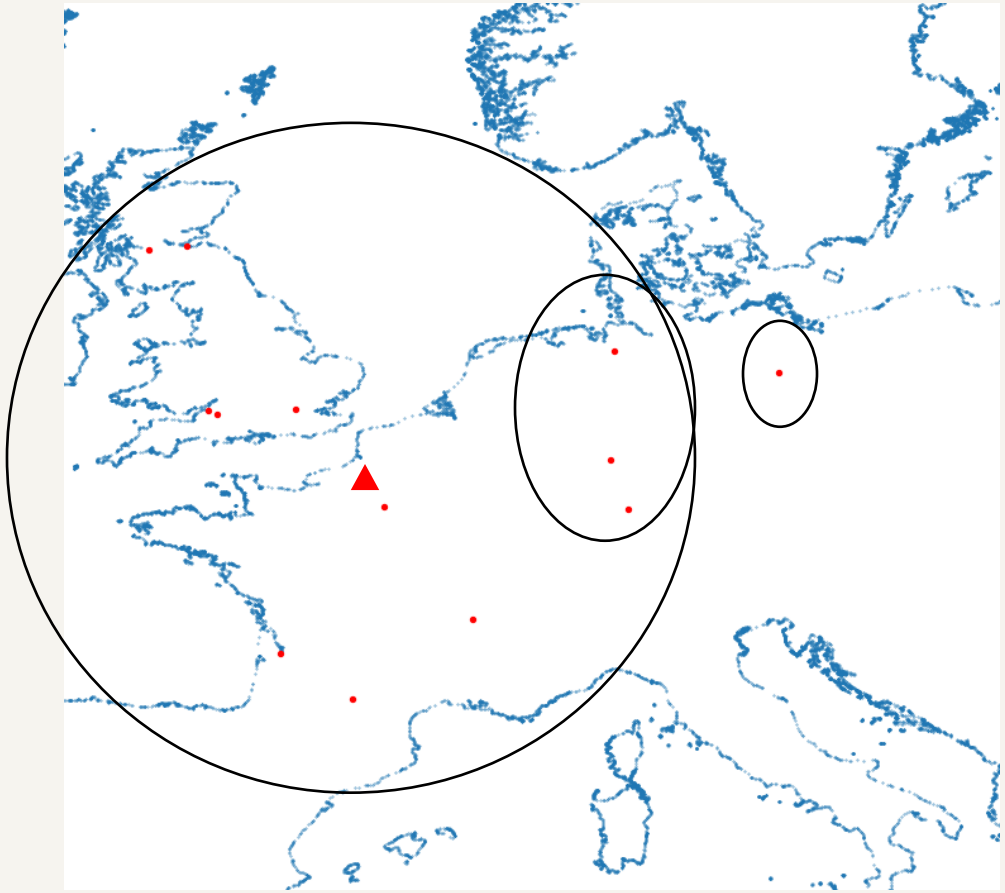
Divisive Clustering Example

- Calculate the distance of the datapoints to the centre of the cluster
- Identify the most remote datapoint
- Use it to define a new cluster



Divisive Clustering Example

- Calculate the distance of the datapoints to the centre of the cluster
- Identify the most remote datapoint
- Use it to define a new cluster
- Calculate the distance of the datapoints to the centres of the two clusters
- Find the datapoints that are closer to the new cluster



Divisive Clustering Example

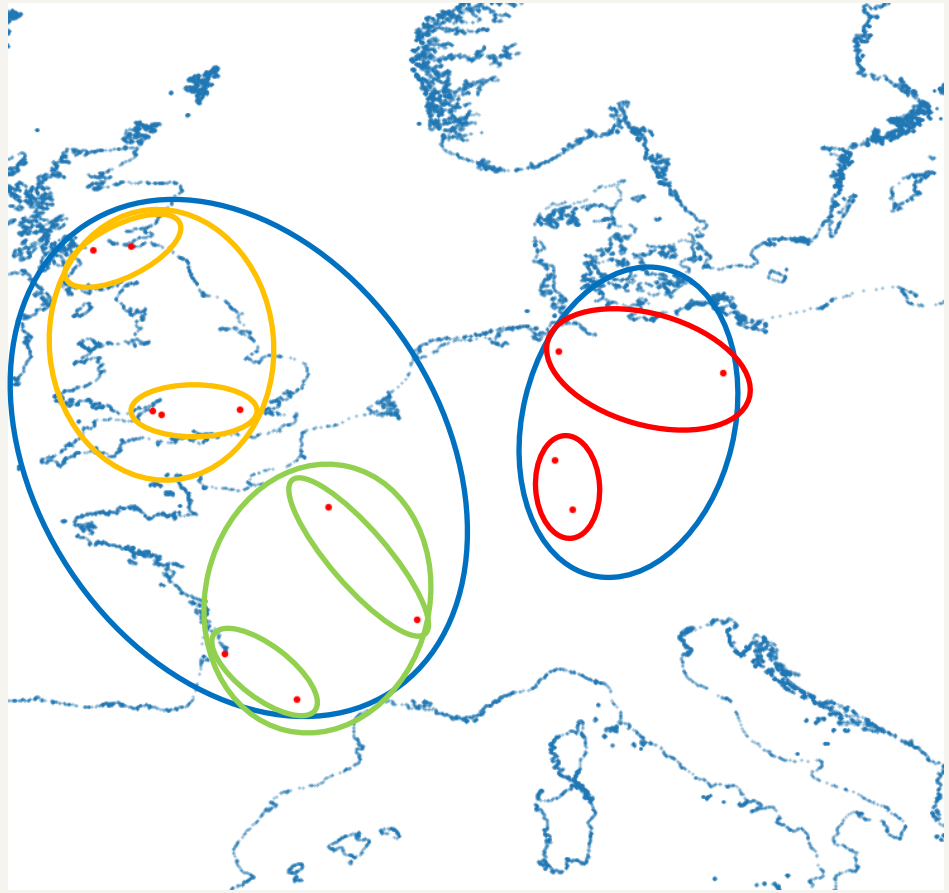
- Calculate the distance of the datapoints to the centre of the cluster
- Identify the most remote datapoint
- Use it to define a new cluster
- Calculate the distance of the datapoints to the centres of the two clusters
- Find the datapoints that are closer to the new cluster
- Move them to the new cluster

Divisive Clustering Example

- Calculate the distance of the datapoints to the centre of the cluster
- Identify the most remote datapoint
- Use it to define a new cluster
- Calculate the distance of the datapoints to the centres of the two clusters
- Find the datapoints that are closer to the new cluster
- Move them to the new cluster
- Repeat the process until you have assigned each datapoint to each one cluster

Divisive Clustering Example

- **Simpler (and more popular) Approach**
- Iteratively Split the Dataset in two, e.g. with **k-means**
- It is possible to choose the number of clusters or continue until all the clusters contain only one datapoint



Comparison of Divisive and Agglomerative

Agglomerative

Complexity in general is $O(N^3)$ but with some implementation assumptions it can be reduced to $O(N^2 \log_2 N)$. All to all distances are required in agglomerative clustering so complexity is bounded from bellow by $\mathrm{O}(N^2)$.

More **flexible** in defining distances and cluster linkage criteria

Easier to implement

Best for fine detail in the data

Divisive

Complexity in a worse case scenario is $O(N^3)$ but with a K-Means based it reduces to $O(\bar{T}\, N \log_2 N)$, where $\bar{T}$ is the average number of K-Means iterations. For most cases $\bar{T} \ll N$.

More difficult to implement

Better for large datasets with clear cluster separation.

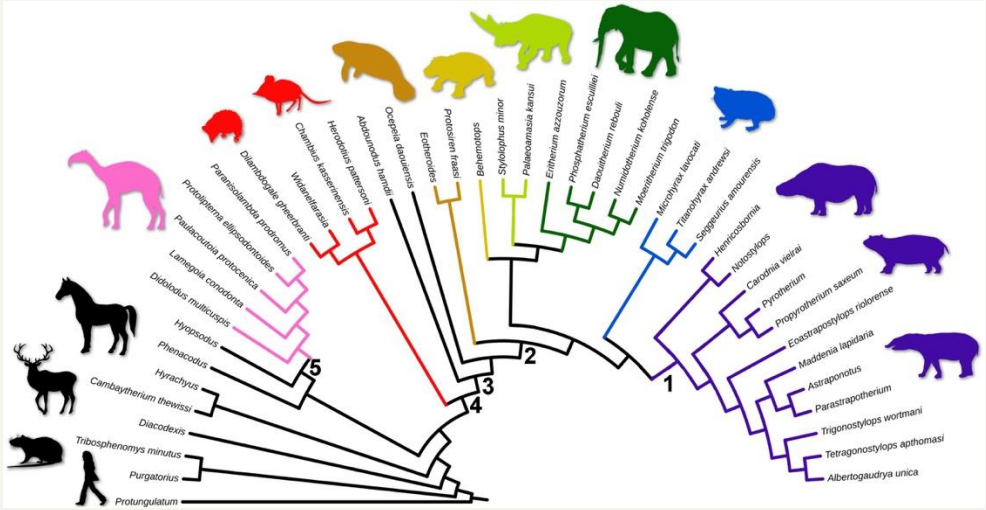
Comparison of Divisive and Agglomerative

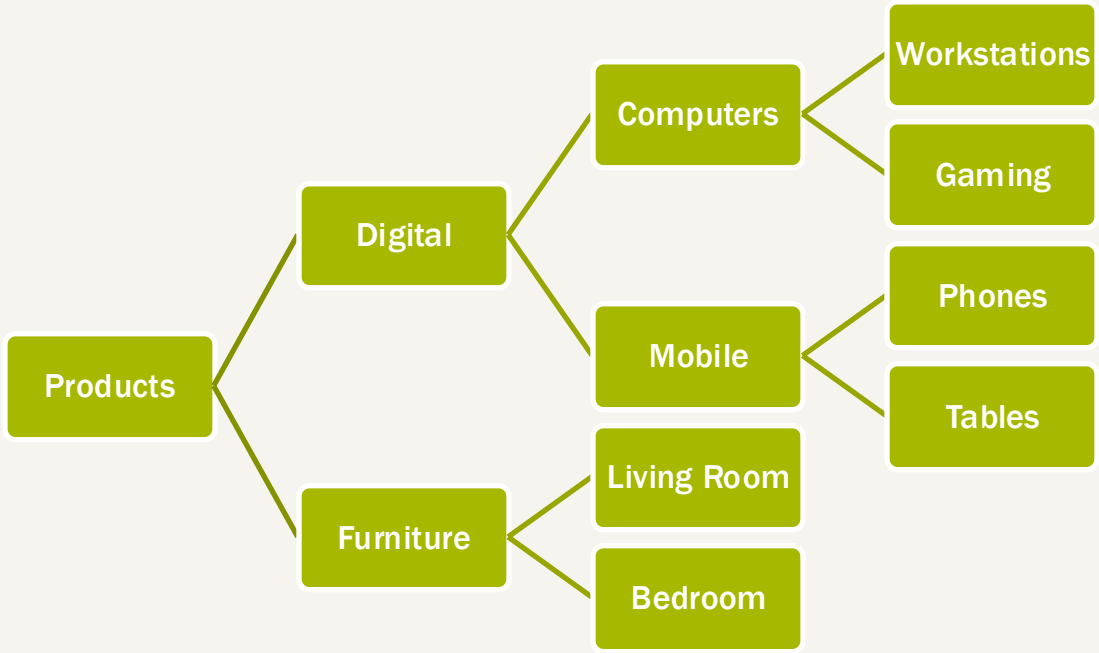
Agglomerative

- A few thousand samples
- Very detailed relationship (specific gene expressions)

Divisive

- Product Categories on a website
- Millions of Products
- Clearly separate categories, e.g. tags





Questions?

Summary

Agglomerative clustering:
- Bottom-up hierarchy
- Linkage
- Few datapoints/Accurate comparison

Divisive Clustering:
- Top-down hierarchy
- Many datapoints/Rough comparison

Thank you!

Questions?

CM22009 Semester 1 Anonymous Feedback Form 2025/26